

CloudFront is most suitable here and is designed to do this.

upvoted 1 times

PaulGa 8 months ago

Selected Answer: C

Ans C - Cloudfront is designed to do this...

upvoted 2 times

TariqKipkemei 1 year, 9 months ago

Selected Answer: C

Amazon CloudFront to the rescue

upvoted 2 times

cookieMr 1 year, 11 months ago

Selected Answer: C

CloudFront is a content delivery network (CDN) service provided by AWS. It caches content at edge locations worldwide, allowing users to access the content quickly regardless of their geographic location. By connecting the S3 to CloudFront, the media files can be cached at edge locations, ensuring reliable and fast delivery to users.

A. is a data transfer service that is not designed for caching or content delivery. It is used for transferring data between on-premises storage systems and AWS services.

B. is a service that improves the performance and availability of applications for global users. While it can provide fast and reliable access, it is not specifically designed for caching media files or connecting directly to S3.

D. is a message queue service that is not suitable for caching or content delivery. It is used for decoupling and coordinating message-based communication between different components of an application.

Therefore, the correct solution is option C, deploying CloudFront to connect the S3 to CloudFront edge servers.

upvoted 5 times

jackky3123213 1 year, 12 months ago

Global Accelerator does not support Edge Caching

upvoted 1 times

Bmarodi 2 years ago

Selected Answer: C

Option C is correct answer.

upvoted 1 times

warioverde 2 years, 2 months ago

As far as I understand, Global Accelerator does not have caching features, so CloudFront would be the recommended service for that purpose

upvoted 2 times

Americo32 2 years, 3 months ago

Selected Answer: C

C correto

upvoted 1 times

ProfXsamson 2 years, 4 months ago

C, Caching == Edge location == CloudFront

upvoted 3 times

career360guru 2 years, 5 months ago

Selected Answer: C

C right answer

upvoted 2 times

L1kai1 2 years, 6 months ago

KIKAVII 2 years, 6 months ago

Selected Answer: C

Agreed

upvoted 1 times

Wpcorgan 2 years, 6 months ago

C is correct

upvoted 1 times

MyNamelsJulien 2 years, 7 months ago

Selected Answer: C

Answer is C

upvoted 1 times

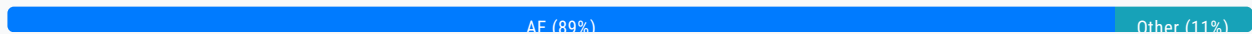
Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #156

Topic 1

A company produces batch data that comes from different databases. The company also produces live stream data from network sensors and application APIs. The company needs to consolidate all the data into one place for business analytics. The company needs to process the incoming data and then stage the data in different Amazon S3 buckets. Teams will later run one-time queries and import the data into a business intelligence tool to show key performance indicators (KPIs). Which combination of steps will meet these requirements with the LEAST operational overhead? (Choose two.)

- A. Use Amazon Athena for one-time queries. Use Amazon QuickSight to create dashboards for KPIs. **Most Voted**
- B. Use Amazon Kinesis Data Analytics for one-time queries. Use Amazon QuickSight to create dashboards for KPIs.
- C. Create custom AWS Lambda functions to move the individual records from the databases to an Amazon Redshift cluster.
- D. Use an AWS Glue extract, transform, and load (ETL) job to convert the data into JSON format. Load the data into multiple Amazon OpenSearch Service (Amazon Elasticsearch Service) clusters.
- E. Use blueprints in AWS Lake Formation to identify the data that can be ingested into a data lake. Use AWS Glue to crawl the source, extract the data, and load the data into Amazon S3 in Apache Parquet format. **Most Voted**

Correct Answer: AE*Community vote distribution***Comments****Wazhija** **Highly Voted** 2 years, 7 months ago**Selected Answer: AE**

I believe AE makes the most sense
upvoted 17 times

Six_Fingered_Jose **Highly Voted** 2 years, 7 months ago**Selected Answer: AE**

yeah AE makes sense, only E is working with S3 here and questions wants them to be in S3
upvoted 12 times

Dharmarajan **Most Recent** 4 months, 1 week ago**Selected Answer: AE**

A&E - least operational overhead - E over D
upvoted 1 times

sdelena 4 months, 2 weeks ago

Selected Answer: AE

A,E is ok
upvoted 1 times

PaulGa 8 months ago

Selected Answer: AD

Ans A, D - A everyone seems to agree; I choose D over E because Parquet is aimed at columnar data - and that is not specified and may restrict query type access
upvoted 2 times

jaradat02 10 months, 3 weeks ago

Selected Answer: AE

AE satisfies the requirements that demand that the data should be stored in s3 and a one-time analytic will run on it.
upvoted 3 times

lofzee 1 year ago

Selected Answer: AE

C and D = too much overhead
B = incorrect because Athena is used for one time queries.

That leaves A and E
upvoted 3 times

awsgeek75 1 year, 5 months ago

Selected Answer: AE

A is a given due to Athena and QuickSight option.
Between C and E, the AWS Lake Formation is a more managed solution so it should have less operational overhead than writing Custom AWS Lambda.
AE should be preferred over AC.
upvoted 4 times

awsgeek75 1 year, 4 months ago

E is only confusing because of Apache Parquet format (like a grid?) what's the point of that in the context of this question?
upvoted 5 times

Guru4Cloud 1 year, 10 months ago

Selected Answer: AE

The reasons are:

AWS Lake Formation and Glue provide automated data lake creation with minimal coding. Glue crawlers identify sources and ETL jobs load to S3.
Athena allows ad-hoc queries directly on S3 data with no infrastructure to manage.
QuickSight provides easy cloud BI for dashboards.
Options C and D require significant custom coding for ETL and queries.
Redshift and OpenSearch would require additional setup and management overhead.
upvoted 8 times

Mia2009687 1 year, 11 months ago

Selected Answer: AE

It combines data from database and stream data, so data lake needs to be used.
And it wants to do one time query, so Athena is better.
upvoted 4 times

TTaws 1 year, 11 months ago

@Golcha once the data comes from different sources then you use GLUE

C moves the data to Redshift while the question says you want it in S3 (and Athena from answer A also needs it in S3).

upvoted 3 times

Jeeva28 2 years ago

Selected Answer: AC

Less Overhead with option AC .No need to manage

upvoted 1 times

pentium75 1 year, 5 months ago

But C moves the data to Redshift while the question says you want it in S3 (and Athena from answer A also needs it in S3).

upvoted 4 times

Golcha 2 years, 2 months ago

Selected Answer: AC

No specific use case for GLUE

upvoted 1 times

TTaws 1 year, 11 months ago

once the data comes from different sources then you use GLUE

upvoted 3 times

pentium75 1 year, 5 months ago

C moves the data to Redshift while the question says you want it in S3 (and Athena from answer A also needs it in S3).

upvoted 2 times

TECHNOWARRIOR 2 years, 2 months ago

The Apache Parquet format is a performance-oriented, column-based data format designed for storage and retrieval. It is generally faster for reads than writes because of its columnar storage layout and a pre-computed schema that is written with the data into the files. AWS Glue's Parquet writer offers fast write performance and flexibility to handle evolving datasets. You can use AWS Glue to read Parquet files from Amazon S3 and from streaming sources as well as write Parquet files to Amazon S3. When using AWS Glue to build a data lake foundation, it automatically crawls your Amazon S3 data, identifies data formats, and then suggests schemas for use with other AWS analytic services[1][2][3][4].

upvoted 6 times

TECHNOWARRIOR 2 years, 2 months ago

ANSWER - AE:Amazon Athena is the best choice for running one-time queries on streaming data. Although Amazon Kinesis Data Analytics provides an easy and familiar standard SQL language to analyze streaming data in real-time, it is designed for continuous queries rather than one-time queries[1]. On the other hand, Amazon Athena is a serverless interactive query service that allows querying data in Amazon S3 using SQL. It is optimized for ad-hoc querying and is ideal for running one-time queries on streaming data[2].AWS Lake Formation uses as a central place to have all your data for analytics purposes (E). Athena integrate perfect with S3 and can makes queries (A).

upvoted 5 times

jcramos 2 years, 2 months ago

Selected Answer: AE

AWS Lake Formation uses as a central place to have all your data for analytics purposes (E). Athena integrate perfect with S3 and can makes queries (A).

upvoted 5 times

jcramos 2 years, 2 months ago

Why S3 in Apache Parquet? <https://aws.amazon.com/about-aws/whats-new/2018/12/amazon-s3-announces-parquet-output-format-for-inventory/>

upvoted 2 times

JiyuKim 2 years, 4 months ago

Can anyone please explain me why B cannot be an answer?

upvoted 6 times

Shrestwt 2 years, 1 month ago

Kinesis Data Analytics is designed for continuous queries rather than one-time queries.
upvoted 8 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

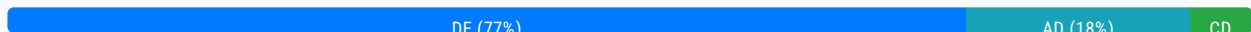
Question #157

Topic 1

A company stores data in an Amazon Aurora PostgreSQL DB cluster. The company must store all the data for 5 years and must delete all the data after 5 years. The company also must indefinitely keep audit logs of actions that are performed within the database. Currently, the company has automated backups configured for Aurora.

Which combination of steps should a solutions architect take to meet these requirements? (Choose two.)

- A. Take a manual snapshot of the DB cluster.
- B. Create a lifecycle policy for the automated backups.
- C. Configure automated backup retention for 5 years.
- D. Configure an Amazon CloudWatch Logs export for the DB cluster. **Most Voted**
- E. Use AWS Backup to take the backups and to keep the backups for 5 years. **Most Voted**

Correct Answer: DE*Community vote distribution***Comments****JayBee65** **Highly Voted** 2 years, 5 months ago

I tend to agree D and E...

A - Manual task that can be automated, so why make life difficult?
B - The maximum retention period is 35 days, so would not help
C - The maximum retention period is 35 days, so would not help
D - Only option that deals with logs, so makes sense
E - Partially manual but only option that achieves the 5 year goal
upvoted 52 times

aadityaravi8 1 year, 11 months ago

100% agree
upvoted 7 times

kmaneith **Highly Voted** 2 years, 6 months ago**Selected Answer: DE**

dude trust me
upvoted 22 times

bcd6d36 1 month ago

I trust you
upvoted 1 times

JayBee65 2 years, 5 months ago

No, please show your reasoning, you may be wrong. Remember, no one thinks they are wrong, but some always are :)
upvoted 16 times

Priyanshugpt486 1 year, 8 months ago

hehe... hehe
upvoted 2 times

PaulGa Most Recent 8 months ago

Selected Answer: DE

Ans D, E
upvoted 1 times

lofzee 1 year ago

D + E I think.
D deals with the logs.
AWS Backup for backups....
Why make backups more difficult by not using the built in backup tool?
upvoted 3 times

rjjkc 1 year ago

Selected Answer: AD

I'd say "A" over "E" because in the option "E", it says use AWS Backup to take the "backups" not snapshot. "If you use AWS CLI, this is set using the parameter DeleteAfterDays. The retention period for snapshots can range between 1 day and 100 years (or indefinitely if you don't enter one), while the retention period for continuous backups can range from 1 day to 35 days. The creation date of a backup is the date the backup job started, not the date it completed. If your backup job doesn't complete on the same date it started, use the date on which it began to help calculate retention periods."

From here: <https://docs.aws.amazon.com/aws-backup/latest/devguide/creating-a-backup-plan.html>
upvoted 2 times

ManikRoy 1 year, 1 month ago

Selected Answer: DE

Agree with the reasoning of JayBee
upvoted 1 times

EMPERBACH 1 year, 1 month ago

Selected Answer: CD

C - C -
<https://www.bing.com/ck/a?!&&p=7e852c106834b9bdJmltdHM9MTcxMzM5ODQwMCZpZ3VpZD0xNzJiMThiOS1kM2RiLTZlZGtEtMWNhZC0wYjRlZDZmZDYmaW5zaWQ9NTQ0OQ&ptn=3&ver=2&hsh=3&fclid=172b18b9-d3db-6eda-1cad-0b4ed2bd6fd6&psq=does+Aurora+offer+auto+backup&u=a1aHR0cHM6Ly9kb2NzLmF3cy5hbWF6b24uY29tL0FtYXpvcjJEUy9sYXRlc3QvQXVyb3JhVXNlckd1aWRlL0F1cm9yYS5NYW5hZ2luZy5CYWNrdXBzLmh0bWwjOn46dGV4dD1BdXJvcmlMjBiYWNRcyUyMHVwJTlweW91ciUyMGNsdXN0ZXlIMjB2b2x1bWUIMjBhdXRvbWFOaWNhbGx5JTlwYW5kLHRvJTlwYW55JTlwcG9pbnQIMjB3aXRoZW4IMjB0aGUIMjBiYWNRdXAlMjByZXRIbnRpb24IMjBwZXJpb2Qu&ntb=1>

D - Only answer for audit logs of activities on database
upvoted 2 times

jjcode 1 year, 4 months ago

My thoughts:
1. AWS backups is designed to make backups
2. "configure backup retention for 5 years" with what? a script? maybe AWS backups???? are the backups done with DD and

2. configure backup retention for 5 years with what? a script? maybe AWS backups???? are the back ups done with DD and stored in S3? i cannot trust this answer
3. "Take a manual snapshot of the DB cluster" this is not an amazon best practice they want us to use their tools AWS backups
4. "create a life cycle policy" assuming the back ups are stored in S3 (which is not a best practice) cannot trust this

that leaves D and E

upvoted 2 times

Ruffyyit 1 year, 6 months ago

D AND E- makes more sense as we automate backups in Aurora DB

- Export data to CloudWatch to capture all log events and configure CloudWatch to retain logs indefinitely.

upvoted 2 times

awashenko 1 year, 8 months ago

Selected Answer: DE

D and E.

A would work as well, but D is the better option as its automated.

E is the only option that gets you to the 5 year retention.

upvoted 2 times

kambarami 1 year, 9 months ago

D AND E- makes more sense as we automate backups in Aurora DB

- Export data to CloudWatch to capture all log events and configure CloudWatch to retain logs indefinitely.

upvoted 2 times

TariqKipkemei 1 year, 9 months ago

Selected Answer: DE

DE makes more sense

upvoted 1 times

Guru4Cloud 1 year, 10 months ago

Selected Answer: CD

The reasons are:

Configuring the automated backups for the Aurora PostgreSQL DB cluster to retain backups for 5 years will meet the requirement to store all data for that duration.

Exporting the database logs to CloudWatch Logs will capture the audit logs of actions performed in the database. CloudWatch Logs retention can be configured to store logs indefinitely.

This meets the need to keep audit logs available beyond the 5 year data retention period.

Additional manual snapshots or using AWS Backup for backups is not necessary since automated backups are already enabled.

A lifecycle policy is useful for transitioning storage classes but does not apply here for a set 5 year retention.

upvoted 2 times

neverdie 2 years, 2 months ago

Selected Answer: AD

Automated backup is limited 35 days

upvoted 3 times

Training4aBetterLife 2 years, 4 months ago

Selected Answer: DE

Previously, you had to create custom scripts to automate backup scheduling, enforce retention policies, or consolidate backup activity for manual Aurora cluster snapshots, especially when coordinating backups across AWS services. With AWS Backup, you gain a fully managed, policy-based backup solution with snapshot scheduling and snapshot retention management. You can now create, manage, and restore Aurora backups directly from the AWS Backup console for both PostgreSQL-compatible and MySQL-compatible versions of Aurora.

To get started, select an Amazon Aurora cluster from the AWS Backup console and take an on-demand backup or simply assign the cluster to a backup plan.

upvoted 5 times

Training4aBetterLife 2 years, 4 months ago

<https://aws.amazon.com/about-aws/whats-new/2020/06/amazon-aurora-snapshots-can-be-managed-via-aws-backup/>

<https://aws.amazon.com/about-aws/whats-new/2020/06/amazon-aurora-snapshots-can-be-managed-via-aws-backup/> : nc1=h_ls

upvoted 3 times

Zerotn3 2 years, 5 months ago

Selected Answer: DE

A is not a valid option for meeting the requirements. A manual snapshot of the DB cluster is a point-in-time copy of the data in the cluster. While taking manual snapshots can be useful for creating backups of the data, it is not a reliable or efficient way to meet the requirement of storing all the data for 5 years and deleting it after 5 years. It would be difficult to ensure that manual snapshots are taken regularly and retained for the required period of time. It is recommended to use a fully managed backup service like AWS Backup, which can automate and centralize the process of taking and retaining backups.

upvoted 4 times

Zerotn3 2 years, 5 months ago

Sorry, B and E that correct

B. Create a lifecycle policy for the automated backups.

This would ensure that the backups taken using AWS Backup are retained for the desired period of time.

upvoted 1 times

JayBee65 2 years, 5 months ago

I think a lifecycle policy would only keep backups for 35 days

upvoted 3 times

awashenko 1 year, 8 months ago

Thats not correct (i thought it was but I went and looked it up) Aurora only keeps backups from 1-35 days.

upvoted 2 times

techhb 2 years, 5 months ago

Selected Answer: DE

D and E only

upvoted 2 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #158

Topic 1

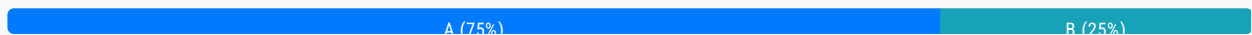
A solutions architect is optimizing a website for an upcoming musical event. Videos of the performances will be streamed in real time and then will be available on demand. The event is expected to attract a global online audience.

Which service will improve the performance of both the real-time and on-demand streaming?

- A. Amazon CloudFront **Most Voted**
- B. AWS Global Accelerator
- C. Amazon Route 53
- D. Amazon S3 Transfer Acceleration

Correct Answer: A

Community vote distribution



Comments

Nigma **Highly Voted** 2 years, 7 months ago

A is right

You can use CloudFront to deliver video on demand (VOD) or live streaming video using any HTTP origin

Global Accelerator is a good fit for non-HTTP use cases, such as gaming (UDP), IoT (MQTT), or Voice over IP, as well as for HTTP use cases that specifically require static IP addresses

upvoted 45 times

mr123dd **Highly Voted** 1 year, 5 months ago

Selected Answer: A

website = http = cloudfront, if it is UDP, then global accelerator

upvoted 12 times

bora4motion 2 weeks, 4 days ago

I thought that a live video stream is UDP.

upvoted 1 times

Rhomerian **Most Recent** 4 months, 1 week ago

Unarmarajan 4 months, 1 week ago

Selected Answer: A

A because CloudFront can be used for video streaming. NOT GA because GA is good for services with lower level network protocols - and their speed ups.

upvoted 1 times

PaulGa 8 months ago

Selected Answer: A

Ans A - CloudFront delivers video on demand or live streaming video using any HTTP origin

upvoted 2 times

dragongoseki 11 months, 3 weeks ago

Selected Answer: A

A is right answer.

upvoted 1 times

ChymKuBoy 11 months, 4 weeks ago

Selected Answer: A

A for sure

upvoted 1 times

trinh_le 12 months ago

AD is correct. link here <https://docs.aws.amazon.com/aws-backup/latest/devguide/creating-a-backup-plan.html>

upvoted 1 times

lofzee 1 year ago

Selected Answer: A

A.

<https://docs.aws.amazon.com/AmazonCloudFront/latest/DeveloperGuide/on-demand-streaming-video.html>

upvoted 2 times

ManikRoy 1 year, 1 month ago

Selected Answer: A

use CloudFront to deliver video on demand (VOD) or live streaming video using any HTTP origin

upvoted 2 times

lanceshen 1 year, 3 months ago

Selected Answer: B

Option A (Amazon CloudFront) is a content delivery network (CDN) service that can improve the performance of on-demand streaming by caching and delivering content from edge locations. While it can accelerate on-demand streaming, it may not provide the same level of optimization for real-time streaming as AWS Global Accelerator.

upvoted 3 times

MrPCarrot 1 year, 4 months ago

A is pperfect <https://d1.awsstatic.com/whitepapers/amazon-cloudfront-for-media.pdf>

upvoted 3 times

awsgeek75 1 year, 5 months ago

Selected Answer: A

<https://docs.aws.amazon.com/AmazonCloudFront/latest/DeveloperGuide/on-demand-streaming-video.html>

CloudFront solves the problem of streaming over Amazon CDN on global scale.

"B" AWS Global Accelerator won't be suitable for streaming from web server as it does not provide edge caching like CDN. Global Accelerator only points the user to nearest functioning node which is helpful for real-time streaming but not best for on-demand.

upvoted 2 times

awsgeek75 1 year, 5 months ago

viru 1 year, 5 months ago

Selected Answer: A

<https://aws.amazon.com/cloudfront/streaming/>
upvoted 3 times

yayaayzo 1 year, 6 months ago

A IS THE RIGHT ANS

. Amazon CloudFront is a content delivery network (CDN) service offered by AWS. It is designed to deliver data, including videos and other media files, with low latency and high transfer speeds. This is a suitable option for optimizing website performance, especially for streaming content globally.

upvoted 2 times

MiniYang 1 year, 6 months ago

Selected Answer: B

Although CloudFront is a content delivery network (CDN) that can provide low-latency and high-performance content delivery, its performance for real-time streaming and on-demand streaming may not be as professional as AWS Global Accelerator

upvoted 1 times

awsgeek75 1 year, 4 months ago

<https://aws.amazon.com/global-accelerator/>

"AWS Global Accelerator is a networking service that helps you improve the availability, performance, and security of your public applications. Global Accelerator provides two global static public IPs that act as a fixed entry point to your application endpoints, such as Application Load Balancers, Network Load Balancers, Amazon Elastic Compute Cloud (EC2) instances, and elastic IPs."

It is Anycast which connect user to closest resource on your server like ALB etc or regional services.

CloudFront is CDN and pushes your content to edge locations near the user. This solves all issues of latency and performance

upvoted 1 times

Ruffyit 1 year, 6 months ago

You can use CloudFront to deliver video on demand (VOD) or live streaming video using any HTTP origin

Global Accelerator is a good fit for non-HTTP use cases, such as gaming (UDP), IoT (MQTT), or Voice over IP, as well as for HTTP use cases that specifically require static IP addresses

upvoted 3 times

mhka1988 1 year, 7 months ago

Selected Answer: A

CloudFront offers several options for streaming your media to global viewers—both pre-recorded files and live events.

<https://docs.aws.amazon.com/AmazonCloudFront/latest/DeveloperGuide/IntroductionUseCases.html#IntroductionUseCasesStreaming>

For video on demand (VOD) streaming, you can use CloudFront to stream in common formats such as MPEG DASH, Apple HLS, Microsoft Smooth Streaming, and CMAF, to any device.

For broadcasting a live stream, you can cache media fragments at the edge, so that multiple requests for the manifest file that delivers the fragments in the right order can be combined, to reduce the load on your origin server.

upvoted 4 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #159

Topic 1

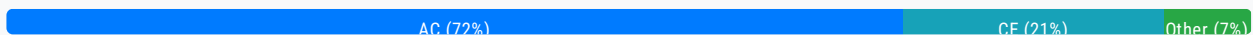
A company is running a publicly accessible serverless application that uses Amazon API Gateway and AWS Lambda. The application's traffic recently spiked due to fraudulent requests from botnets.

Which steps should a solutions architect take to block requests from unauthorized users? (Choose two.)

- A. Create a usage plan with an API key that is shared with genuine users only. **Most Voted**
- B. Integrate logic within the Lambda function to ignore the requests from fraudulent IP addresses.
- C. Implement an AWS WAF rule to target malicious requests and trigger actions to filter them out. **Most Voted**
- D. Convert the existing public API to a private API. Update the DNS records to redirect users to the new API endpoint.
- E. Create an IAM role for each user attempting to access the API. A user will assume the role when making the API call.

Correct Answer: AC

Community vote distribution



Comments

jdr75 **Highly Voted** 2 years, 2 months ago

Selected Answer: CE

C) WAF has bot identification and remedial tools, so it's CORRECT.

A) remember the question : "...block requests from unauthorized users?" -- an api key is involved in a authorization process. It's not the more secure process, but it's better than an totoally anonymous process. If you don't know the key, you can't authenticate. So the bots, at least the first days/weeks could not access the service (at the end they'll do, cos' the key will be spread informally). So it's CORRECT.

B) Implement a logic in the Lambda to detect fraudulent ip's is almost impossible, cos' it's a dynamic and changing pattern that you cannot handle easily.

D) creating a rol is not going to imply be more protected from unauth. request, because a rol is a "principal", it's not involved in the authorization process.

upvoted 10 times

debasishdtt 1 year, 5 months ago

Don't use API keys for authentication or authorization to control access to your APIs. If you have multiple APIs in a usage plan

Don't use API keys for authentication or authorization to control access to your APIs. If you have multiple APIs in a usage plan, a user with a valid API key for one API in that usage plan can access all APIs in that usage plan. Instead, to control access to your API, use an IAM role, a Lambda authorizer, or an Amazon Cognito user pool.

upvoted 2 times

pentium75 1 year, 5 months ago

E "An IAM role for EACH (!) user ATTEMPTING (!) to access the API"? Hello no.

upvoted 5 times

awashenko Highly Voted 1 year, 8 months ago

Selected Answer: AC

Agree A and C

I don't see how E is feasible as its a public API. How would you create an IAM role for each user?

upvoted 6 times

Mrigraj12 Most Recent 4 months, 1 week ago

Selected Answer: AC

Wrong choices: B-> Integrating logic in lambda is not feasible as you will have to feed data which will increase its size, execution time and therefore overall cost.

D-> If it happens then the public will not be able to access it as it is private only clients inside VPC will be able to access it.

E-> You are just giving them access

upvoted 1 times

PaulGa 8 months ago

Selected Answer: AC

Ans A, C -

A: using API keys and usage plans restricts access to your API to users who have the key, limiting fraudulent access.

C: designed to fight bots

upvoted 2 times

diddy99 11 months, 3 weeks ago

Selected Answer: AC

C) Everyone agrees on C

B) Almost impossible cos how do you detect fraudulent IP address from a publicly accessible application

D) It's a publicly accessible application, converting the API to a private one defeats the purpose

E) IAM role for each user trying to access a publicly accessible API is impossible. It like creating an IAM for each user that tries to use google AUTH for their website

A) By implementing API keys and usage plans, you can restrict access to your API to only those users who possess the key, helping to limit fraudulent access.

upvoted 2 times

jerry1 2 months, 3 weeks ago

but i think both option A and E imply that the api access actually have a targeted user
if A is possible, why cant E be an approach?

upvoted 1 times

dragongoseki 11 months, 3 weeks ago

Selected Answer: AC

AC is right answer.

upvoted 1 times

ChymKuBoy 11 months, 4 weeks ago

Selected Answer: AC

AC for sure

upvoted 2 times

hb0011 1 year ago

Do the people voting E realize how insane that is? Creating a local IAM user in your account for every user that needs to access the API. No just... no.

upvoted 3 times

EMPERBACH 1 year, 1 month ago

Selected Answer: BC

B. Integrate logic within the Lambda function to ignore the requests from fraudulent IP addresses. -> you can think about CORS script write on Lambda to prevent fraudulent IP addresses.

C. Implement an AWS WAF rule to target malicious requests and trigger actions to filter them out. -> No comment here as it can use to filter traffic

upvoted 1 times

MrPCarrot 1 year, 4 months ago

C and D are the perfect answers

upvoted 2 times

debasishdtta 1 year, 5 months ago

Don't use API keys for authentication or authorization to control access to your APIs. If you have multiple APIs in a usage plan, a user with a valid API key for one API in that usage plan can access all APIs in that usage plan. Instead, to control access to your API, use an IAM role, a Lambda authorizer, or an Amazon Cognito user pool.

upvoted 1 times

awsgeek75 1 year, 5 months ago

Selected Answer: CD

I'll throw a curveball over here. "C" is a given as WAF rules can target malicious usage. For example:

<https://docs.aws.amazon.com/whitepapers/latest/aws-best-practices-ddos-resiliency/aws-waf-ip-reputation.html>

"D" Convert existing public API to a private API. This part is same as A. The additional bit over here is to change the DNS record to a new API endpoint which blocks the requests from unauthorised users also. The unauthorised users will not be redirected from public to private API endpoint. I am assuming that the public API endpoint will be used for authorisation and only authorised users will be redirected to private endpoint. This is more robust as the actual API (private endpoint) never gets hit with requests from unauthorised bots and WAF redirects it back to public URL.

Happy to be corrected and challenged

upvoted 4 times

sidharthwader 1 year, 3 months ago

It's a globally published API if you make it private how do other people access it ? A would be the better solution than D

upvoted 2 times

ale_brd_111 1 year, 6 months ago

Selected Answer: AC

The combination of using an API key and implementing an AWS WAF rule provides the most comprehensive and effective way to block requests from unauthorized users and protect the company's serverless application from botnet attacks.

upvoted 4 times

MiniYang 1 year, 6 months ago

Selected Answer: CE

A. Create plans using API keys shared only with real users: While using API keys is a standard way to control access to APIs, using API keys alone may not completely prevent attacks from botnets. Malicious request.

B. Incorporate logic in the Lambda function to ignore requests from fraudulent IP addresses: This may be a solution, but filtering that relies more on IP addresses may not be as flexible as using AWS WAF.

D. Convert an existing public API to a private API. Update DNS records to redirect users to the new API endpoint: This approach makes the API private, but requires user redirects and may inconvenience existing users.

upvoted 1 times

Ruffytit 1 year, 6 months ago

C) WAF has bot identification and remedial tools, so it's CORRECT.

A) remember the question : "...block requests from unauthorized users?" -- an api key is involved in a authorization process. It's not the more secure process, but it's better than an totoally anonymous process. If you don't know the key, you can't authenticate. So the bots, at least the first days/weeks could not access the service (at the end they'll do, cos' the key will be spread informally). So it's CORRECT.

upvoted 2 times

TariqKipkemei 1 year, 9 months ago

Selected Answer: AC

AWS WAF rule to target and filter out malicious requests and API key to authorize users.

upvoted 1 times

Guru4Cloud 1 year, 10 months ago

Selected Answer: AC

The reasons are:

An API key with a usage plan limits access to only authorized apps and users. This prevents general public access. WAF rules can identify and block malicious bot traffic through pattern matching and IP reputation lists. Together, the API key and WAF provide preventative and detective controls against unauthorized requests. The other options add complexity or are reactive. IAM roles per user is not feasible for a public API. Ignoring requests in Lambda and changing DNS are response actions after an attack.

upvoted 3 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #160

Topic 1

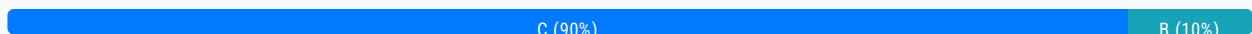
An ecommerce company hosts its analytics application in the AWS Cloud. The application generates about 300 MB of data each month. The data is stored in JSON format. The company is evaluating a disaster recovery solution to back up the data. The data must be accessible in milliseconds if it is needed, and the data must be kept for 30 days.

Which solution meets these requirements MOST cost-effectively?

- A. Amazon OpenSearch Service (Amazon Elasticsearch Service)
- B. Amazon S3 Glacier
- C. Amazon S3 Standard **Most Voted**
- D. Amazon RDS for PostgreSQL

Correct Answer: C

Community vote distribution

**Comments**

babaxoxo **Highly Voted** 2 years, 6 months ago

Selected Answer: C

Ans C:

Cost-effective solution with milliseconds of retrieval -> it should be s3 standard
upvoted 14 times

Guru4Cloud **Highly Voted** 1 year, 10 months ago

Selected Answer: C

The reasons are:

S3 Standard provides high durability and availability for storage

It allows millisecond access to retrieve objects

Objects can be stored for any duration, meeting the 30 day retention need

Storage costs are low, around \$0.023 per GB/month

OpenSearch and RDS require running and managing a cluster for DR storage

Glacier has lower cost but retrieval time is too high at 3-5 hours

S3 Standard's simplicity, high speed access, and low cost make it optimal for this small DR dataset that needs to be accessed quickly

upvoted 6 times

satyaamm Most Recent 4 months, 1 week ago

Selected Answer: C

S3 Standard is the most cheapest.

upvoted 1 times

PaulGa 7 months, 4 weeks ago

Selected Answer: C

Ans C - S3 std: cost-effective, milliseconds of retrieval

upvoted 2 times

PaulGa 8 months ago

Selected Answer: C

S3 Standard

upvoted 1 times

soufiyane 1 year, 2 months ago

Selected Answer: C

also s3 can store any form of data

upvoted 1 times

pentium75 1 year, 5 months ago

Selected Answer: C

Only Glacier class that would meet the requirement is Instant Retrieval, but it has 90 days minimum storage time which would kill the cost savings.

upvoted 5 times

xdkonorek2 1 year, 7 months ago

Selected Answer: B

300 MB / month storage without retrieval when file is single 300 MB file:

S3 Standard cost (Monthly): 0.01 USD

S3 Standard cost (Upfront): 0.00 USD

S3 Glacier Instant Retrieval cost (Monthly): 0.00 USD

if it was 3GB:

3 GB / month storage without retrieval when file is single 3GB file:

S3 Standard cost (Monthly): 0.07 USD

S3 Standard cost (Upfront): 0.00 USD

S3 Glacier Instant Retrieval cost (Monthly): 0.01 USD

When assumed no retrieval is required because it's DR solution, and it's a single file, Glacier Instant Retrieval wins, and when they mention S3 glacier we must choose one of the sub-category

upvoted 1 times

xdkonorek2 1 year, 7 months ago

but if 300 MB is divided into smaller files situation changes which is probably the case..

300 MB / month storage without retrieval when files are 600x0.5MB :

S3 Standard cost (Monthly): 0.01 USD

S3 Standard cost (Upfront): 0.00 USD

S3 Glacier Instant Retrieval cost (Monthly): 0.01 USD

S3 Glacier Instant Retrieval cost (Upfront): 0.02 USD

3 GB / month storage without retrieval when files are 6000x0.5 MB file:

S3 Standard cost (Monthly): 0.10 USD

S3 Standard cost (Upfront): 0.03 USD

S3 Glacier Instant Retrieval cost (Monthly): 0.13 USD

S3 Glacier Instant Retrieval cost (Upfront): 0.25 USD

upvoted 4 times

pentium75 1 year, 5 months ago

Only Glacier Instant Retrieval (which is not mentioned in B) would meet the access requirement, but "Objects that are archived to S3 Glacier Instant Retrieval and S3 Glacier Flexible Retrieval are charged for a minimum storage duration of 90 days."
upvoted 2 times

Its_SaKar 1 year, 9 months ago

Selected Answer: C

Answer is not B because S3 glacier and S3 glacier instant storage are two different types of storage class. So, answer here is C: S3 standard
upvoted 4 times

TariqKipkemei 1 year, 9 months ago

Selected Answer: C

Data must be accessible in milliseconds and must be kept for 30 days = Amazon S3 Standard
upvoted 2 times

chanchal133 1 year, 9 months ago

Selected Answer: C

ANS - C
upvoted 1 times

Nazmul123 1 year, 10 months ago

Selected Answer: C

<https://aws.amazon.com/s3/storage-classes/glacier/instant-retrieval/>
upvoted 2 times

pentium75 1 year, 5 months ago

Not mentioned in B, and 90 days minimum storage time.
upvoted 1 times

cookieMr 1 year, 11 months ago

S3 Standard is a highly durable and scalable storage option suitable for backup and disaster recovery purposes. It offers millisecond access to data when needed and provides durability guarantees. It is also cost-effective compared to other storage options like OpenSearch Service, S3 Glacier, and RDS for PostgreSQL, which may have higher costs or longer access times for retrieving the data.

A. OpenSearch Service (Elasticsearch Service): While it offers fast data retrieval, it may incur higher costs compared to storing data directly in S3, especially considering the amount of data being generated.

B. S3 Glacier: While it provides long-term archival storage at a lower cost, it does not meet the requirement of immediate access in milliseconds. Retrieving data from Glacier typically takes several hours.

D. RDS for PostgreSQL: While it can be used for data storage, it may be overkill and more expensive for a backup and disaster recovery solution compared to S3 Standard, which is more suitable and cost-effective for storing and retrieving data.
upvoted 3 times

joehong 2 years ago

Selected Answer: B

<https://aws.amazon.com/s3/storage-classes/glacier/instant-retrieval/>
upvoted 4 times

pentium75 1 year, 5 months ago

"Objects that are archived to S3 Glacier Instant Retrieval and S3 Glacier Flexible Retrieval are charged for a minimum storage duration of 90 days." Also Instant Retrieval is not mentioned in B.
upvoted 2 times

KZM 2 years, 4 months ago

A. Incorrect

Amazon OpenSearch Service (Amazon Elasticsearch Service) is designed for full-text search and analytics, but it may not be the most cost-effective solution for this use case

B. Incorrect

S3 Glacier is a cold storage solution that is designed for long-term data retention and infrequent access.

C. Correct

S3 standard is cost-effective and meets the requirement. S3 Standard allows for data retention for a specific number of days.

D. PostgreSQL is a relational database service and may not be the most cost-effective solution.

upvoted 5 times

ngochieu276 2 years, 5 months ago

Selected Answer: B

S3 Glacier Instant Retrieval – Use for archiving data that is rarely accessed and requires milliseconds retrieval.

<https://docs.aws.amazon.com/amazonglacier/latest/dev/introduction.html>

upvoted 4 times

career360guru 2 years, 5 months ago

Selected Answer: C

Option C

upvoted 1 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

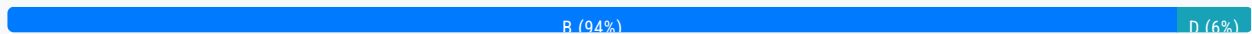
Question #161

Topic 1

A company has a small Python application that processes JSON documents and outputs the results to an on-premises SQL database. The application runs thousands of times each day. The company wants to move the application to the AWS Cloud. The company needs a highly available solution that maximizes scalability and minimizes operational overhead.

Which solution will meet these requirements?

- A. Place the JSON documents in an Amazon S3 bucket. Run the Python code on multiple Amazon EC2 instances to process the documents. Store the results in an Amazon Aurora DB cluster.
- B. Place the JSON documents in an Amazon S3 bucket. Create an AWS Lambda function that runs the Python code to process the documents as they arrive in the S3 bucket. Store the results in an Amazon Aurora DB cluster. **Most Voted**
- C. Place the JSON documents in an Amazon Elastic Block Store (Amazon EBS) volume. Use the EBS Multi-Attach feature to attach the volume to multiple Amazon EC2 instances. Run the Python code on the EC2 instances to process the documents. Store the results on an Amazon RDS DB instance.
- D. Place the JSON documents in an Amazon Simple Queue Service (Amazon SQS) queue as messages. Deploy the Python code as a container on an Amazon Elastic Container Service (Amazon ECS) cluster that is configured with the Amazon EC2 launch type. Use the container to process the SQS messages. Store the results on an Amazon RDS DB instance.

Correct Answer: B*Community vote distribution***Comments****babaxoxo** **Highly Voted** 2 years, 6 months ago**Selected Answer: B**

solution should remove operation overhead -> s3 -> lambda -> aurora
upvoted 18 times

markw92 1 year, 12 months ago

Aurora supports mysql and postgresql but question has database sql server. So, that eliminates B. So, the other logical answer is D. IMHO. Btw, i also thought the answer is B and started re-reading question carefully.
upvoted 6 times

JIJIXI 1 year, 8 months ago

sql database, not sql server
upvoted 6 times

Zerotn3 Highly Voted 2 years, 5 months ago

Selected Answer: B

By placing the JSON documents in an S3 bucket, the documents will be stored in a highly durable and scalable object storage service. The use of AWS Lambda allows the company to run their Python code to process the documents as they arrive in the S3 bucket without having to worry about the underlying infrastructure. This also allows for horizontal scalability, as AWS Lambda will automatically scale the number of instances of the function based on the incoming rate of requests. The results can be stored in an Amazon Aurora DB cluster, which is a fully-managed, high-performance database service that is compatible with MySQL and PostgreSQL. This will provide the necessary durability and scalability for the results of the processing.

upvoted 13 times

satyaamm Most Recent 4 months ago

Selected Answer: B

Using Lambda, S3 and Aurora is the most suitable and provides least operational overhead for this problem.

upvoted 1 times

PaulGa 7 months, 4 weeks ago

Selected Answer: D

Ans D - because none of the other answers guarantee FIFO. What if a subsequent JSON operation is intended to update an existing database record (which is highly likely) - then the wrong change would be applied

upvoted 2 times

soufiyane 1 year, 2 months ago

Selected Answer: B

b is the right answer it very obvious

upvoted 1 times

Anantvir 1 year, 5 months ago

Guys I have a question.

We dont know how long the processing of JSON documents is going to take. What if that processing takes more than 15 min ? Lambda can run only for 15 correct ? Based on this the answer could be D

Please correct my understanding.

upvoted 3 times

awsgeek75 1 year, 5 months ago

Selected Answer: B

"D" has a lot of moving parts and more operational overhead even if each part is a managed service in itself. Also, if something can be done with Lambda, don't use an EC2 instance in any form as it always increases operational overhead (compared to Lambda).

upvoted 3 times

David_Ang 1 year, 7 months ago

Selected Answer: B

"D" is just like the most complex one, sometimes the admin make mistakes and don't realize. lambda is a service make for this

upvoted 2 times

Mandar15 1 year, 8 months ago

Selected Answer: B

B is correc

upvoted 1 times

TariqKipkemei 1 year, 9 months ago

Selected Answer: B

Main requirement is: 'scalability and minimized operational overhead' = serverless = Amazon S3 bucket, AWS Lambda function, Amazon Aurora DB cluster

upvoted 2 times

Guru4Cloud 1 year, 10 months ago

Selected Answer: B

- Using Lambda functions triggered by S3 events allows the Python code to automatically scale up and down based on the number of incoming JSON documents. This provides high availability and maximizes scalability.
- Storing the results in an Amazon Aurora DB cluster provides a managed, scalable, and highly available database.
- This serverless approach minimizes operational overhead since Lambda and Aurora handle provisioning infrastructure, deploying code, monitoring, patching, etc.

upvoted 3 times

aadityaravi8 1 year, 11 months ago

The answer is B. Place the JSON documents in an Amazon S3 bucket. Create an AWS Lambda function that runs the Python code to process the documents as they arrive in the S3 bucket. Store the results in an Amazon Aurora DB cluster.

This solution is highly available because Lambda functions are automatically scaled up or down based on the number of requests they receive. It is also scalable because you can easily add more Lambda functions to process more documents. Finally, it minimizes operational overhead because you do not need to manage any EC2 instances.

upvoted 2 times

cookieMr 1 year, 11 months ago

Selected Answer: B

Using Lambda eliminates the need to manage and provision servers, ensuring scalability and minimizing operational overhead. S3 provides durable and highly available storage for the JSON documents. Lambda can be triggered automatically whenever new documents are added to the S3 bucket, allowing for real-time processing. Storing the results in an Aurora DB cluster ensures high availability and scalability for the processed data. This solution leverages serverless architecture, allowing for automatic scaling and high availability without the need for managing infrastructure, making it the most suitable choice.

A. This option requires manual management and scaling of EC2 instances, resulting in higher operational overhead and complexity.

C. This approach still involves manual management and scaling of EC2 instances, increasing operational complexity and overhead.

D. This solution requires managing and scaling an ECS cluster, adding operational overhead and complexity. Utilizing SQS adds complexity to the system, requiring custom handling of message consumption and processing in the Python code.

upvoted 4 times

Bmarodi 2 years ago

Selected Answer: B

Keywords here are : "maximizes scalability and minimizes operational overhead, hence option B is correct answer.

upvoted 2 times

channn 2 years, 2 months ago

Selected Answer: D

i vote for D as 'on-premises SQL database' is not mysql/postgre which can replace by aurora

upvoted 2 times

pentium75 1 year, 5 months ago

Why not? It's a "SQL database", NOT necessarily Microsoft SQL Server. But even if it would be SQL server, that could be migrated to Aurora.

upvoted 1 times

kerin 2 years, 3 months ago

B is the best option. <https://aws.amazon.com/rds/aurora/>

upvoted 1 times

mp165 2 years, 5 months ago

Selected Answer: B

agree...B is the best option S3, Lambda , Aurora.

upvoted 2 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #162

Topic 1

A company wants to use high performance computing (HPC) infrastructure on AWS for financial risk modeling. The company's HPC workloads run on Linux. Each HPC workflow runs on hundreds of Amazon EC2 Spot Instances, is short-lived, and generates thousands of output files that are ultimately stored in persistent storage for analytics and long-term future use.

The company seeks a cloud storage solution that permits the copying of on-premises data to long-term persistent storage to make data available for processing by all EC2 instances. The solution should also be a high performance file system that is integrated with persistent storage to read and write datasets and output files.

Which combination of AWS services meets these requirements?

- A. Amazon FSx for Lustre integrated with Amazon S3 **Most Voted**
- B. Amazon FSx for Windows File Server integrated with Amazon S3
- C. Amazon S3 Glacier integrated with Amazon Elastic Block Store (Amazon EBS)
- D. Amazon S3 bucket with a VPC endpoint integrated with an Amazon Elastic Block Store (Amazon EBS) General Purpose SSD (gp2) volume

Correct Answer: A*Community vote distribution*

A (100%)

Comments**Marge_Simpson** **Highly Voted** 2 years, 6 months ago**Selected Answer: A**

If you see HPC and Linux both in the question.. Pick Amazon FSx for Lustre
upvoted 45 times

HayLLiHuK 2 years, 5 months ago

yeap, you're right!
upvoted 5 times

aba2s **Highly Voted** 2 years, 5 months ago

Selected Answer: A

Additional keywords: make data available for processing by all EC2 instances ==> FSx

In absence of EFS, it should be FSx. Amazon FSx For Lustre provides a high-performance, parallel file system for hot data
upvoted 11 times

satyaamm **Most Recent** 4 months ago

Selected Answer: A

Lustre is the most suitable when HPC is required.
upvoted 1 times

PaulGa 7 months, 4 weeks ago

Selected Answer: A

Ans A - HPC so its FSx and Lustre
upvoted 3 times

MrPCarrot 1 year, 4 months ago

A is the answer because Amazon FSx for Lustre provides a high-performance, scalable file system optimized for compute-intensive workloads like HPC.
upvoted 3 times

awsgeek75 1 year, 5 months ago

Selected Answer: A

Lustre is default when HPC is involved. <https://aws.amazon.com/fsx/lustre/>

B: mentions Windows and no-one asked for it.

C: S3 Glacier is too slow for HPC

D: I don't think this is possible, unless I'm mistake, how can you connect a VPC endpoint to EBS without an EC2 kind of instance?
upvoted 2 times

TariqKipkemei 1 year, 9 months ago

Selected Answer: A

HPC workloads running on Linux = Amazon FSx for Lustre
upvoted 2 times

Jeyaluxshan 1 year, 9 months ago

High performance - Lustre
upvoted 2 times

Guru4Cloud 1 year, 10 months ago

Selected Answer: A

The reasons are:

Amazon FSx for Lustre provides a high-performance, scalable file system optimized for compute-intensive workloads like HPC. It has native integration with Amazon S3.

Data can be copied from on-premises to an S3 bucket, acting as persistent long-term storage.

The FSx for Lustre file system can then access the S3 data for high speed processing of datasets and output files.

FSx for Lustre is designed for the Linux environments used in this HPC workload.

upvoted 3 times

cookieMr 1 year, 11 months ago

Selected Answer: A

FSx for Lustre is a high-performance file system optimized for compute-intensive workloads. It provides scalable, parallel access to data and is suitable for HPC applications.

By integrating FSx for Lustre with S3, you can easily copy on-premises data to long-term persistent storage in S3, making it available for processing by EC2 instances.

S3 serves as the durable and highly scalable object storage for storing the output files, allowing for analytics and long-term future use.

Option B, FSx for Windows File Server, is not suitable because the workloads run on Linux, and this option is designed for

Windows file sharing.

Option C, S3 Glacier integrated with EBS, is not the best choice as it is a low-cost archival storage service and not optimized for high-performance file system requirements.

Option D, using an S3 bucket with a VPC endpoint integrated with an Amazon EBS General Purpose SSD (gp2) volume, does not provide the required high-performance file system capabilities for HPC workloads.

upvoted 5 times

Bmarodi 2 years ago

Selected Answer: A

Option A is right answer.

upvoted 1 times

kerin 2 years, 3 months ago

FSx for Lustre makes it easy and cost-effective to launch and run the popular, high-performance Lustre file system. You use Lustre for workloads where speed matters, such as machine learning, high performance computing (HPC), video processing, and financial modeling.

Amazon FSx for Lustre is integrated with Amazon S3.

upvoted 3 times

SilentMilli 2 years, 5 months ago

Selected Answer: A

Amazon FSx for Lustre integrated with Amazon S3

upvoted 2 times

techhb 2 years, 5 months ago

Selected Answer: A

A is right choice here.

upvoted 1 times

career360guru 2 years, 5 months ago

Selected Answer: A

Option A is the best high performance storage with integration to S3

upvoted 2 times

wly_al 2 years, 6 months ago

Selected Answer: A

requirement is File System and workload running on linux. so S3 and FSx for windows is not an option

upvoted 2 times

Shasha1 2 years, 6 months ago

A
The Amazon FSx for Lustre service is a fully managed, high-performance file system that makes it easy to move and process large amounts of data quickly and cost-effectively. It provides a fully managed, cloud-native file system with low operational overhead, designed for massively parallel processing and high-performance workloads. The Lustre file system is a popular, open source parallel file system that is well-suited for a variety of applications such as HPC, image processing, AI/ML, media processing, data analytics, and financial modeling, among others. With Amazon FSx for Lustre, you can quickly create and configure new file systems in minutes, and easily scale the size of your file system up or down

upvoted 3 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #163

Topic 1

A company is building a containerized application on premises and decides to move the application to AWS. The application will have thousands of users soon after it is deployed. The company is unsure how to manage the deployment of containers at scale. The company needs to deploy the containerized application in a highly available architecture that minimizes operational overhead.

Which solution will meet these requirements?

- A. Store container images in an Amazon Elastic Container Registry (Amazon ECR) repository. Use an Amazon Elastic Container Service (Amazon ECS) cluster with the AWS Fargate launch type to run the containers. Use target tracking to scale automatically based on demand. **Most Voted**
- B. Store container images in an Amazon Elastic Container Registry (Amazon ECR) repository. Use an Amazon Elastic Container Service (Amazon ECS) cluster with the Amazon EC2 launch type to run the containers. Use target tracking to scale automatically based on demand.
- C. Store container images in a repository that runs on an Amazon EC2 instance. Run the containers on EC2 instances that are spread across multiple Availability Zones. Monitor the average CPU utilization in Amazon CloudWatch. Launch new EC2 instances as needed.
- D. Create an Amazon EC2 Amazon Machine Image (AMI) that contains the container image. Launch EC2 instances in an Auto Scaling group across multiple Availability Zones. Use an Amazon CloudWatch alarm to scale out EC2 instances when the average CPU utilization threshold is breached.

Correct Answer: A

Community vote distribution

A (100%)

Comments

goatbernard **Highly Voted** 2 years, 7 months ago

Selected Answer: A

AWS Fargate
upvoted 17 times

awsgeek75 **Highly Voted** 1 year, 5 months ago

Selected Answer: A

A is minimal overhead.
B has EC2 overhead
C EC2 instance overhead + container repository running on EC2 overhead
D AMII, CloudWatch alarm is overhead++
upvoted 8 times

awsgeek75 1 year, 4 months ago

Also, simply speaking, if the company is unsure how to manage deployed containers then Fargate is the only choice.
upvoted 2 times

satyaamm **Most Recent** 4 months ago

Selected Answer: A

AWS Fargate provides least operational overhead.
upvoted 1 times

PaulGa 7 months, 3 weeks ago

Selected Answer: A

Ans A - AWS Fargate - it does it all
upvoted 2 times

lofzee 1 year ago

Selected Answer: A

fargate is the compute managed for you.. A
upvoted 2 times

Ruffyit 1 year, 6 months ago

ECR+ECS+Fargate = Less overhead
upvoted 4 times

ACloud_Guru15 1 year, 7 months ago

Selected Answer: A

ECR+ECS+Fargate = Less overhead
upvoted 3 times

Sindokuhlep 1 year, 7 months ago

Selected Answer: A

Fargate
upvoted 2 times

TariqKipkemei 1 year, 9 months ago

Selected Answer: A

Highly available architecture that minimizes operational overhead = Serverless = Elastic Container Registry, Amazon Elastic Container Service with AWS Fargate launch type
upvoted 2 times

Guru4Cloud 1 year, 10 months ago

Selected Answer: A

Using ECR provides a fully managed container image registry.
ECS with Fargate launch type allows running containers without managing servers or clusters. Fargate will handle scaling and optimization.
Target tracking autoscaling will allow automatically adjusting capacity based on demand.
The serverless approach with Fargate minimizes operational overhead.
upvoted 3 times

MikeDu 1 year, 10 months ago

Selected Answer: A

AWF Fargate should be the best choice
upvoted 2 times

aadityaravi8 1 year, 11 months ago

A is the right answer undoubtedly.
upvoted 1 times

cookieMr 1 year, 11 months ago

Selected Answer: A

ECR provides a secure and scalable repository to store and manage container images. ECS with the Fargate launch type allows you to run containers without managing the underlying infrastructure, providing a serverless experience. Target tracking in ECS can automatically scale the number of tasks or services based on a target value such as CPU or memory utilization, ensuring that the application can handle increasing demand without manual intervention.

Option B is not the best choice because using the EC2 launch type requires managing and scaling EC2 instances, which increases operational overhead.

Option C is not the optimal solution as it involves managing the container repository on an EC2 instance and manually launching EC2 instances, which adds complexity and operational overhead.

Option D also requires managing EC2 instances, configuring ASGs, and setting up manual scaling rules based on CloudWatch alarms, which is not as efficient or scalable as using Fargate in combination with ECS.

upvoted 6 times

Bmarodi 1 year, 11 months ago

Nice explanations!
upvoted 1 times

Bmarodi 2 years ago

Selected Answer: A

ECS + Fargate satisfy requirements, hence option A is the best solution.

upvoted 2 times

studynoplay 2 years, 1 month ago

Selected Answer: A

minimize operational overhead = Serverless

Fargate is Serverless

upvoted 2 times

NoInNothing 2 years, 2 months ago

Selected Answer: A

Correct is "A"

upvoted 1 times

jaswantn 2 years, 2 months ago

You can place Fargate launch type all in one AZ, or across multiple AZs. But Option A does not take care of High Availability requirement of question. With Option C we have multi AZ.

upvoted 2 times

wizcloudifa 1 year, 3 months ago

that was my doubt too, is Fargate by default highly available? I chose D as option C didn't have a scalability option in it, option D has an autoscaling group in it C doesn't

upvoted 1 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #164

Topic 1

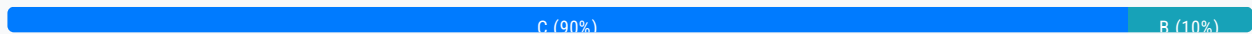
A company has two applications: a sender application that sends messages with payloads to be processed and a processing application intended to receive the messages with payloads. The company wants to implement an AWS service to handle messages between the two applications. The sender application can send about 1,000 messages each hour. The messages may take up to 2 days to be processed. If the messages fail to process, they must be retained so that they do not impact the processing of any remaining messages.

Which solution meets these requirements and is the MOST operationally efficient?

- A. Set up an Amazon EC2 instance running a Redis database. Configure both applications to use the instance. Store, process, and delete the messages, respectively.
- B. Use an Amazon Kinesis data stream to receive the messages from the sender application. Integrate the processing application with the Kinesis Client Library (KCL).
- C. Integrate the sender and processor applications with an Amazon Simple Queue Service (Amazon SQS) queue. Configure a dead-letter queue to collect the messages that failed to process. **Most Voted**
- D. Subscribe the processing application to an Amazon Simple Notification Service (Amazon SNS) topic to receive notifications to process. Integrate the sender application to write to the SNS topic.

Correct Answer: C

Community vote distribution

**Comments**

aba2s **Highly Voted** 2 years, 5 months ago

Selected Answer: C

Amazon SQS supports dead-letter queues (DLQ), which other queues (source queues) can target for messages that can't be processed (consumed) successfully.

<https://docs.aws.amazon.com/AWSSimpleQueueService/latest/SQSDeveloperGuide/sqs-dead-letter-queues.html>
upvoted 13 times

cookieMr **Highly Voted** 1 year, 11 months ago

Selected Answer: C

By integrating both the sender and processor applications with an SQS, messages can be reliably sent from the sender to the processor application for processing. SQS provides at-least-once delivery, ensuring that messages are not lost in transit. If a message fails to process, it can be retained in the queue and retried without impacting the processing of other messages. Configuring a DLQ allows for the collection of messages that repeatedly fail to process, providing visibility into failed messages for troubleshooting and analysis.

A is not the optimal choice as it involves managing and configuring an EC2 instance running a Redis, which adds operational overhead and maintenance requirements.

B is not the most operationally efficient solution as it introduces additional complexity by using Amazon Kinesis data streams and integrating with the Kinesis Client Library for message processing.

D, using SNS, is not the best fit for the scenario as it is more suitable for pub/sub messaging and broadcasting notifications rather than the specific requirement of message processing between two applications.

upvoted 7 times

Bmarodi 1 year, 11 months ago

Nice explanations always, thanks a lot!

upvoted 1 times

Bmarodi 1 year, 10 months ago

Nice explanations always, thanks a lot

upvoted 1 times

satyaamm Most Recent 4 months ago

Selected Answer: C

Dead Letter queues are the most suitable here.

upvoted 1 times

PaulGa 7 months, 3 weeks ago

Selected Answer: B

Ans C - Use SQS and dead letter queue.

Not ideal because max visibility timeout for SQS is 12h and can't be extended by user.

I did consider B: Kinesis which is for real-time processing - altho' the question doesn't say it's not real-time. Anyway, SQS is basic and fits.

upvoted 2 times

dkw2342 1 year, 3 months ago

C) This option works. There is a 12h maximum visibility timeout, but:

"If you don't know how long it takes to process a message, create a heartbeat for your consumer process: Specify the initial visibility timeout (for example, 2 minutes) and then—as long as your consumer still works on the message—keep extending the visibility timeout by 2 minutes every minute."

<https://docs.aws.amazon.com/AWSSimpleQueueService/latest/SQSDeveloperGuide/working-with-messages.html>

upvoted 4 times

dkw2342 1 year, 3 months ago

None of the options fit.

A) Not operationally efficient

B) Kinesis is for real-time processing

D) SNS is not suitable for work queuing.

C) While this may be the "correct" answer, it also doesn't really fit the problem statement.

Maximum visibility timeout for SQS is 12h, also can't be extended by the consumer.

"If your consumer needs longer than 12 hours, consider using Step Functions."

<https://docs.aws.amazon.com/AWSSimpleQueueService/latest/SQSDeveloperGuide/sqs-visibility-timeout.html>

upvoted 2 times

MatAlves 9 months, 2 weeks ago

It doesn't seem to be the case though:

" For example, assume that a message spends 1 day in the original queue before it's moved to a dead-letter queue. If the dead-letter queue's retention period is 4 days, the message is deleted from the dead-letter queue after 3 days and the ApproximateAgeOfOldestMessage is 3 days."

<https://docs.aws.amazon.com/AWSSimpleQueueService/latest/SQSDeveloperGuide/sqs-dead-letter-queues.html#understanding-message-retention-periods>

upvoted 2 times

awsgeek75 1 year, 5 months ago

Selected Answer: C

C is the only option with dead letter que which meets the requirement of retaining messages that fail to process without impacting other messages.

upvoted 3 times

TariqKipkemei 1 year, 9 months ago

Selected Answer: C

Implement an AWS service to handle messages between the two applications = Amazon Simple Queue Service
If the messages fail to process, they must be retained = a dead-letter queue

upvoted 4 times

Guru4Cloud 1 year, 10 months ago

Selected Answer: C

SQS provides a fully managed message queuing service that meets all the requirements:

SQS can handle the sending and processing of 1,000 messages per hour
Messages can be retained for up to 14 days to allow the full 2 days for processing
Using a dead-letter queue will retain failed messages without impacting other processing
SQS requires minimal operational overhead compared to running your own message queue server

upvoted 3 times

MutiverseAgent 1 year, 10 months ago

Selected Answer: B

Answer is B), the reason is:

- Because messages might up to 2 days to be processed. Visibility timeout of SQS is 12 hours, so after 12 hours another consumer might take a message from the queue which is currently being processed.

upvoted 1 times

Jeeva28 2 years ago

Selected Answer: C

Answer C, In Question if Keyword have Processing Failed >> SQS

upvoted 2 times

Bmarodi 2 years ago

Selected Answer: C

solution that meets these requirements and is the MOST operationally efficient will be option C. SQS is buffer between 2 APPs.

upvoted 2 times

norris81 2 years ago

The visibility timeout must not be more than 12 hours. (For SQS)

Jobs may take 2 days to process

upvoted 2 times

studynoplay 2 years, 1 month ago

Selected Answer: C

operationally efficient = Serverless
SQS is serverless

SQS is serverless

upvoted 2 times

studynoplay 2 years, 1 month ago

SNS too is serverless, but it is obvious that it is not the correct answer in this case

upvoted 1 times

apchandana 2 years, 1 month ago

Selected Answer: C

more realistic option is C.

only problem with this is the limit of the visibility timeout is 12H max. as the second application take 2 days to process, there will be a duplicate of processing messages in the queue. this might complicate things.

upvoted 3 times

nilandd44gg 1 year, 11 months ago

Amazon SQS automatically deletes messages that have been in a queue for more than the maximum message retention period. The default message retention period is 4 days. However, you can set the message retention period to a value from 60 seconds to 1,209,600 seconds (14 days) using the SetQueueAttributes action.

<https://docs.aws.amazon.com/AWSSimpleQueueService/latest/SQSDeveloperGuide/sqs-basic-architecture.html>

upvoted 2 times

vherman 2 years, 3 months ago

SQS has a limit 12h for visibility time out

upvoted 1 times

bullrem 2 years, 4 months ago

Selected Answer: B

Option C, using Amazon SQS, is a valid solution that meets the requirements of the company. However, it may not be the most operationally efficient solution because SQS is a managed message queue service that requires additional operational overhead to handle the retention of messages that failed to process. Option B, using Amazon Kinesis Data Streams, is more operationally efficient for this use case because it can handle the retention of messages that failed to process automatically and provides the ability to process and analyze streaming data in real-time.

upvoted 1 times

UnluckyDucky 2 years, 3 months ago

Kinesis stream save data for up to 24 hours, doesn't meet the 2 day requirement.

Kinesis streams don't have fail-safe for failed processing, unlike SQS.

The correct answer is C - SQS.

upvoted 4 times

apchandana 2 years, 1 month ago

this is not a correct statement.

A data stream is a logical grouping of shards. There are no bounds on the number of shards within a data stream (request a limit increase if you need more). A data stream will retain data for 24 hours by default, or optionally up to 365 days.

Shard

<https://aws.amazon.com/kinesis/data-streams/getting-started/>

upvoted 1 times

LuckyAro 2 years, 4 months ago

There's no way for kinesis to know whether the message processing failed.

upvoted 1 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

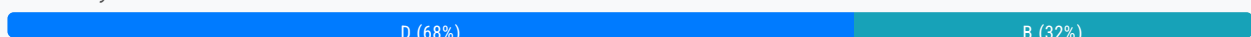
Question #165

Topic 1

A solutions architect must design a solution that uses Amazon CloudFront with an Amazon S3 origin to store a static website. The company's security policy requires that all website traffic be inspected by AWS WAF.

How should the solutions architect comply with these requirements?

- A. Configure an S3 bucket policy to accept requests coming from the AWS WAF Amazon Resource Name (ARN) only.
- B. Configure Amazon CloudFront to forward all incoming requests to AWS WAF before requesting content from the S3 origin.
- C. Configure a security group that allows Amazon CloudFront IP addresses to access Amazon S3 only. Associate AWS WAF to CloudFront.
- D. Configure Amazon CloudFront and Amazon S3 to use an origin access identity (OAI) to restrict access to the S3 bucket. Enable AWS WAF on the distribution. **Most Voted**

Correct Answer: D*Community vote distribution***Comments****Nigma** **Highly Voted** 2 years, 7 months ago

Answer D. Use an OAI to lockdown CloudFront to S3 origin & enable WAF on CF distribution
upvoted 35 times

FNJ1111 2 years, 5 months ago

<https://aws.amazon.com/premiumsupport/knowledge-center/cloudfront-access-to-amazon-s3/> confirms use of OAI (and option D).
upvoted 14 times

cookieMr **Highly Voted** 1 year, 11 months ago**Selected Answer: B**

By configuring CloudFront to forward all incoming requests to AWS WAF, the traffic will be inspected by AWS WAF before reaching the S3 origin, complying with the security policy requirement. This approach ensures that all website traffic is inspected by AWS WAF, providing an additional layer of security before accessing the content stored in the S3 origin.

Option A is not the correct choice as configuring an S3 bucket policy to accept requests from the AWS WAF ARN only would bypass the inspection of traffic by AWS WAF. It does not ensure that all website traffic is inspected.

Option C is not the optimal solution as it focuses on controlling access to S3 using a security group. Although it associates AWS WAF with CloudFront, it does not guarantee that all incoming requests are inspected by AWS WAF.

Option D is not the recommended solution as configuring an OAI in CloudFront and restricting access to the S3 bucket does not ensure that all website traffic is inspected by AWS WAF. The OAI is used for restricting direct access to S3 content, but the traffic should still pass through AWS WAF for inspection.

upvoted 11 times

Tsige 7 months, 3 weeks ago

CloudFront does not "forward" requests to AWS WAF. Instead, AWS WAF integrates directly with CloudFront to inspect traffic as it passes through the distribution. There is no manual forwarding process involved.

so, my Answer is D.

upvoted 2 times

escalibran 1 year, 3 months ago

Option B does use the WAF through Cloudfront, but it does not mention anything to prevent direct access to the objects without going through Cloudfront.

upvoted 3 times

bogobob 1 year, 7 months ago

Apparently you can only point to a custom host that is "not an Amazon Simple Storage Service (Amazon S3) bucket" (other than for static hosting). <https://aws.amazon.com/blogs/security/how-to-enhance-amazon-cloudfront-origin-security-with-aws-waf-and-aws-secrets-manager/>. Answer should be D

upvoted 4 times

SinghJagdeep 1 year, 5 months ago

agreed. Must be D as per above security blog

upvoted 4 times

Vandaman Most Recent 3 months, 3 weeks ago

Selected Answer: B

The requirement is for the website traffic to be inspected by WAF, not to ensure that the S3 bucket is only accessible through CloudFront. Hence, I say B

upvoted 1 times

satyaammm 4 months ago

Selected Answer: D

Using OAI for CloudFront is the most suitable option here.

upvoted 1 times

jayessh 7 months ago

Selected Answer: D

option D is correct

This is the correct approach because it ensures that the S3 bucket is only accessible through CloudFront, and AWS WAF can inspect all incoming traffic to the CloudFront distribution.

why option B is incorrect

because AWS WAF is integrated with CloudFront at the edge locations. CloudFront does not forward requests to AWS WAF; instead, AWS WAF inspects the requests as they come into CloudFront

upvoted 3 times

PaulGa 7 months, 3 weeks ago

Selected Answer: B

Ans B - configure CloudFront to forward incoming requests to AWS WAF for inspection before sending to S3. This provides an additional layer of security before accessing the content stored in the S3 origin.

D: not ideal because configuring an OAI in CloudFront and restricting access to the S3 bucket does not guarantee website traffic is inspected by WAF.

upvoted 1 times

maryam_sh 10 months, 1 week ago

Selected Answer: D

B is incorrect, it misrepresents how AWS WAF works with CloudFront. AWS WAF is not an intermediary service that CloudFront forwards requests to. Instead, AWS WAF is directly integrated with CloudFront as a layer to inspect incoming requests. The correct configuration is to associate AWS WAF with the CloudFront distribution, not to forward requests separately.

upvoted 5 times

jatric 11 months, 1 week ago

Selected Answer: D

Cloud Front allows configuration to enable AWS WAF and restrict direct access to S3 through OAI will meet the requirements.

upvoted 3 times

Duckydoo 11 months, 3 weeks ago

Selected Answer: D

The requirements indicate that S3 is used to "store" a static website, not that it must be configured as a static website (which does not make any sense if it's to be used with CF anyway). Furthermore, the requirements also indicate that all traffic must be inspected by WAF. If you do not setup OAI/OAC, you can potentially bypass CF and access S3 directly. So option B does not satisfy the second requirement.

upvoted 4 times

shil_31 1 year ago

Selected Answer: D

Using an OAI (Origin Access Identity) restricts access to the S3 bucket, ensuring that only CloudFront can access the content. Enabling AWS WAF on the CloudFront distribution allows you to inspect website traffic and filter out malicious requests before they reach your S3 origin.

upvoted 2 times

ManikRoy 1 year, 1 month ago

Selected Answer: D

OAI is required so that S3 bucket is not accessed directly.

upvoted 4 times

Hkayne 1 year, 1 month ago

I guess D

upvoted 1 times

keehua 1 year, 2 months ago

Selected Answer: D

There are two ways you can serve static websites on AWS S3 origin, either using website endpoints or REST API endpoints. Website endpoints does not support HTTPS. Note that the question does not mention which endpoint is used.

<https://docs.aws.amazon.com/AmazonS3/latest/userguide/WebsiteEndpoints.html#WebsiteRestEndpointDiff>

B is incorrect because we do not 'forward' requests to AWS WAF, we attach WAF on the Cloudfront distribution itself. Could be bad wording of the question.

D is totally valid because it doesn't mention using website endpoints. D also uses OAI to restrict direct access to objects in AWS S3. Although OAI is still viable in this point of time, it is still a legacy method and it is more recommended to use OAC instead.

<https://repost.aws/knowledge-center/cloudfront-serve-static-website>

upvoted 4 times

Uzbekistan 1 year, 2 months ago

Selected Answer: B

Option B ensures that all incoming requests to the static website served through Amazon CloudFront are first forwarded to AWS WAF for inspection before the content is requested from the S3 origin. This ensures that all website traffic is inspected by AWS WAF as required by the company's security policy.

upvoted 2 times

drdz13 1 year, 3 months ago

D is not possible since you cannot set OAC or OAI if S3 bucket is used as static website host
upvoted 1 times

bujuman 1 year, 4 months ago

Selected Answer: D

WAF is associated to a Cloudfront Distribution
upvoted 2 times

awsgeek75 1 year, 4 months ago

Selected Answer: D

A: Doesn't make sense in context with CF.

B: You configure WAF on CF for HTTP status handling so this may be right be is badly worded

C: You might as well re-engineer S3 and CloudFront!

D: The requirement for WAF usage is met with this. Doesn't have to be smart usage, just enabled.

upvoted 3 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #166

Topic 1

Organizers for a global event want to put daily reports online as static HTML pages. The pages are expected to generate millions of views from users around the world. The files are stored in an Amazon S3 bucket. A solutions architect has been asked to design an efficient and effective solution.

Which action should the solutions architect take to accomplish this?

- A. Generate presigned URLs for the files.
- B. Use cross-Region replication to all Regions.
- C. Use the geoproximity feature of Amazon Route 53.
- D. Use Amazon CloudFront with the S3 bucket as its origin. **Most Voted**

Correct Answer: D

Community vote distribution

D (100%)

Comments

Buruguduystunstugudunstuy **Highly Voted** 2 years, 5 months ago

Selected Answer: D

The most effective and efficient solution would be Option D (Use Amazon CloudFront with the S3 bucket as its origin.)

Amazon CloudFront is a content delivery network (CDN) that speeds up the delivery of static and dynamic web content, such as HTML pages, images, and videos. By using CloudFront, the HTML pages will be served to users from the edge location that is closest to them, resulting in faster delivery and a better user experience. CloudFront can also handle the high traffic and large number of requests expected for the global event, ensuring that the HTML pages are available and accessible to users around the world.

upvoted 11 times

cookieMr **Highly Voted** 1 year, 11 months ago

Selected Answer: D

CloudFront is well-suited for efficiently serving static HTML pages to users around the world. By using it with the S3 as its origin, the static HTML pages can be cached and distributed globally to edge locations, reducing latency and improving performance for users accessing the pages from different regions. This solution ensures efficient and effective delivery of the daily reports to millions of users worldwide, providing a scalable and high-performance solution for the global event.

A would allow temporary access to the files, but it does not address the scalability and performance requirements of serving millions of views globally.

B is not necessary for this scenario as the goal is to distribute the static HTML pages efficiently to users worldwide, not replicate the files across multiple Regions.

C is primarily used for routing DNS traffic based on the geographic location of users, but it does not provide the caching and content delivery capabilities required for this use case.

upvoted 5 times

satyaamm Most Recent 4 months ago

Selected Answer: D

CloudFront is the most suitable option here.

upvoted 1 times

PaulGa 7 months, 2 weeks ago

Selected Answer: D

Ans D - CloudFront content delivery netwk is designed for the job: speeds delivery of static/ dynamic content (HTML, images, vids). Pages will be served from the closest edge location = faster delivery; improved UX; managed high vol traffic; high vol requests

upvoted 2 times

TariqKipkemei 1 year, 9 months ago

Selected Answer: D

Global users = Amazon CloudFront

upvoted 2 times

Guru4Cloud 1 year, 10 months ago

Selected Answer: D

CloudFront is the best solution for this use case because:

CloudFront is a content delivery network (CDN) that caches content at edge locations around the world. This brings content closer to users for fast performance.

For high traffic global events with millions of viewers, a CDN is necessary for effective distribution.

Using the S3 bucket as the origin, CloudFront can fetch the files once and cache them globally.

upvoted 2 times

career360guru 2 years, 5 months ago

Selected Answer: D

Option D

upvoted 1 times

k1kavi1 2 years, 6 months ago

Selected Answer: D

Agreed

upvoted 1 times

Sahilbhai 2 years, 6 months ago

answer is D agree with Shasha1

upvoted 1 times

Shasha1 2 years, 6 months ago

D
CloudFront is a content delivery network (CDN) offered by Amazon Web Services (AWS). It functions as a reverse proxy service that caches web content across AWS's global data centers, improving loading speeds and reducing the strain on origin servers. CloudFront can be used to efficiently deliver large amounts of static or dynamic content anywhere in the world.

upvoted 3 times

Wpcorgan 2 years, 6 months ago

D is correct

upvoted 2 times

Nigma 2 years, 7 months ago

D

Static content on S3 and hence Cloudfront is the best way

upvoted 2 times

Pamban 2 years, 7 months ago

Selected Answer: D

D is the correct answer

upvoted 2 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #167

Topic 1

A company runs a production application on a fleet of Amazon EC2 instances. The application reads the data from an Amazon SQS queue and processes the messages in parallel. The message volume is unpredictable and often has intermittent traffic. This application should continually process messages without any downtime.

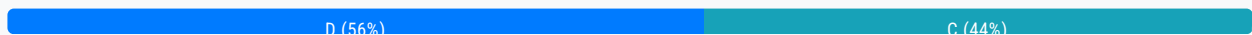
Which solution meets these requirements MOST cost-effectively?

- A. Use Spot Instances exclusively to handle the maximum capacity required.
- B. Use Reserved Instances exclusively to handle the maximum capacity required.
- C. Use Reserved Instances for the baseline capacity and use Spot Instances to handle additional capacity.
- D. Use Reserved Instances for the baseline capacity and use On-Demand Instances to handle additional capacity.

Most Voted

Correct Answer: D

Community vote distribution



Comments

HayLLIHuK Highly Voted 2 years, 5 months ago

Selected Answer: C

"without any downtime" - Reserved Instances for the baseline capacity
 "MOST cost-effectively" - Spot Instances to handle additional capacity
 upvoted 46 times

LuckyAro 2 years, 4 months ago

Dude, read the question, cost consideration was not mentioned in the question.
 upvoted 3 times

ShinobiGrappler 2 years, 4 months ago

Dude, read the question, "Which solution meets these requirements MOST cost-effectively?"
 upvoted 46 times

kraken21 2 years, 2 months ago

kraken21 2 years, 2 months ago

I am leaning towards C because the idea of having a queue is to decouple the processing. If an instance goes down(spot) while processing will it not show up back after the visibility timeout? So using spot meets the cost-effective objective.
upvoted 5 times

Sutariya 1 year, 9 months ago

Intermediate data stored in SQS queue so once free then it take data and process.
upvoted 1 times

MrSaint 2 years, 1 month ago

cost-effectively means, Cheapest solution (cost) that achieve all the requirements (effectively). Its not cost-effectively if is just cheapest solution that fail to address all the requirements, in this case. (This application should continually process messages without any downtime) no matter the volume, since it is unpredictable. B for example, address the requirement but not the cheapest solution that achieve it. D is the cheaper choice that address the requirement (without any downtime). and C is cheaper than D but do not guarantee that you wont have downtime since it is SPOT instances.
upvoted 17 times

kraken21 2 years, 2 months ago

How can you have baseline capacity when your message volume is unpredictable and often has intermittent traffic?
upvoted 4 times

MutiverseAgent 1 year, 10 months ago

For this reason I think correct answer is A
upvoted 1 times

Macadam 1 year, 7 months ago

Spot instances cannot be an option as it is unreliable and the question requires the messages to be continuously processed
upvoted 5 times

taer Highly Voted 2 years, 6 months ago

Selected Answer: D

D is the correct answer
upvoted 28 times

Drayen25 2 years, 4 months ago

C is correct, read for cost effectiveness
upvoted 6 times

diabloexodia 1 year, 11 months ago

AWS has stopped issuing spot instances so i think C
upvoted 1 times

diabloexodia 1 year, 11 months ago

so i think C is incorrect*. the Correct ans is D.
upvoted 2 times

sezer 2 years, 2 months ago

if you cannot find enough spot instance you will have downtime
you cannot always find spot instance
upvoted 13 times

Kumaran1508 2 years ago

Why downtime when there are baseline reserved instances?
upvoted 3 times

creamymanosauce 11 months ago

When traffic spikes and there happens to an increase in the spot price, if the price exceeds the threshold set then there would be downtime as no additional compute can be acquired. Hence even if spot instances are more cost effective, I would interpret the cost effective requirement to be the ability to reliably acquire new instances and dispose of when not needed, hence on demand instances

upvoted 2 times

network_enthusiast Most Recent 1 month, 4 weeks ago

Selected Answer: C

C is the correct answer

upvoted 1 times

Yak_Yeti 2 months ago

Selected Answer: C

"MOST cost-effectively" - Spot Instances to handle additional capacity

The application will process the messages with the reserved instances - slowly - with no downtime.

upvoted 1 times

chest_jd 2 months, 1 week ago

Selected Answer: C

Cost-effective - C.

upvoted 1 times

Faraz999 2 months, 2 weeks ago

Selected Answer: C

I am leaning towards C because the idea of having a queue is to decouple the processing.

upvoted 1 times

triptimandal01 2 months, 2 weeks ago

Selected Answer: C

Spot Instances can be used to handle additional capacity to accommodate intermittent traffic. Spot Instances are ideal for variable and unpredictable workloads because they allow you to take advantage of unused EC2 capacity at a significantly lower cost than On-Demand Instances.

upvoted 1 times

tch 3 months, 1 week ago

Selected Answer: D

baseline capacity" refers to the normal or expected capacity performance of a system or process, so use reserved instance - this to fulfil without any downtime and cost-effective

if "additional capacity" refers to resources or capabilities beyond the standard or baseline capacity needed to handle normal operations, allowing for flexibility and the ability to handle surges in demand or unexpected events - then we should use on-demand instances

upvoted 1 times

iamroyalty_k 4 months ago

Selected Answer: C

C is correct because it balances cost and reliability using a combination of:

- Reserved Instances for steady-state traffic (ensuring continuous processing).
- Spot Instances for handling spikes in traffic (reducing costs significantly).

A - ☐ Not Reliable – Spot Instances can be interrupted at any time. Using them exclusively could cause processing delays or failures.

B - ☐ Not Cost-Effective – RIs are best for predictable workloads, but over-provisioning for peak loads leads to unnecessary costs when traffic is low.

D - ☐ More Expensive Than Spot Instances – On-Demand Instances are more costly than Spot Instances for handling additional capacity.

upvoted 1 times

Dharmarajan 4 months, 1 week ago

Selected Answer: D

I believe the best fit for the scenario is D. D has a portion that is about 90% more expensive than C. However, Spot instances can be interrupted which means that some of the processes can be interrupted, which is not acceptable. So considering that I am choosing D.

upvoted 1 times

Mrigraj12 4 months, 1 week ago

Selected Answer: C

Easily the most cost effective

upvoted 1 times

mayorpasca32 4 months, 1 week ago

Selected Answer: C

Spot Instances – Can be scaled up dynamically when message volume increases, at a much lower cost than On-Demand instances.

upvoted 1 times

skybrink 4 months, 3 weeks ago

Selected Answer: C

Why Option C is Correct:

Reserved Instances: Provide a cost-effective solution for the baseline capacity (predictable, steady traffic) with a significant discount compared to On-Demand Instances.

Spot Instances: Are highly cost-effective (up to 90% cheaper than On-Demand Instances) and can be used for handling additional, variable traffic since the application processes messages in parallel and can tolerate instance interruptions.

Combining these instance types ensures a balance of cost savings and availability.

upvoted 1 times

henrikamirbekyan 5 months, 1 week ago

Selected Answer: C

I think it is C

The instance will remove the message from the queue when it is already processed, there is one downside here when the spot instance is terminated before message removal, it will lead to double processing. The question is not talking about time, so having baseline + spot is the most cost effective here.

upvoted 1 times

HiraPBI 5 months, 3 weeks ago

Selected Answer: C

C - Because of cost effectiveness.

upvoted 1 times

wellzz5422 6 months, 1 week ago

Selected Answer: D

D. Use Reserved Instances for the baseline capacity and use On-Demand Instances to handle additional capacity. Most Voted

upvoted 1 times

tom_cruise 6 months, 2 weeks ago

Selected Answer: D

"without any downtime" rules out spot instance.

upvoted 2 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #168

Topic 1

A security team wants to limit access to specific services or actions in all of the team's AWS accounts. All accounts belong to a large organization in AWS Organizations. The solution must be scalable and there must be a single point where permissions can be maintained.

What should a solutions architect do to accomplish this?

- A. Create an ACL to provide access to the services or actions.
- B. Create a security group to allow accounts and attach it to user groups.
- C. Create cross-account roles in each account to deny access to the services or actions.

D. Create a service control policy in the root organizational unit to deny access to the services or actions. **Most Voted**

Correct Answer: D

Community vote distribution

D (100%)

Comments

Nigma **Highly Voted** 2 years, 7 months ago

D. Service control policies (SCPs) are one type of policy that you can use to manage your organization. SCPs offer central control over the maximum available permissions for all accounts in your organization, allowing you to ensure your accounts stay within your organization's access control guidelines. See

https://docs.aws.amazon.com/organizations/latest/userguide/orgs_manage_policies_scp.html.

upvoted 22 times

cookieMr **Highly Voted** 1 year, 11 months ago

By creating an SCP in the root organizational unit, the security team can define and enforce fine-grained permissions that limit access to specific services or actions across all member accounts. The SCP acts as a guardrail, denying access to specified services or actions, ensuring that the permissions are consistent and applied uniformly across the organization. SCPs are scalable and provide a single point of control for managing permissions, allowing the security team to centrally manage access restrictions without needing to modify individual account settings.

Option A and option B are not suitable for controlling access across multiple accounts in AWS Organizations. ACLs and security groups are typically used for managing network traffic and access within a single account or a specific resource.

Option C is not the recommended approach. Cross-account roles are used for granting access, and denying access through cross-account roles can be complex and less manageable compared to using SCPs.

upvoted 10 times

awashenko 1 year, 8 months ago

This was a good explanation of why A and B are not correct. I was thinking A but after reading this I agree with you D is correct.

upvoted 2 times

satyaamm Most Recent 4 months ago

Selected Answer: D

SCP are most suitable with AWS Organizations

upvoted 1 times

Dharmarajan 4 months, 1 week ago

Selected Answer: D

D is the best fit among the options. Better fit would be using AWS Control tower.

upvoted 1 times

PaulGa 7 months, 2 weeks ago

Selected Answer: D

Ans D - "A service control policy in the root organizational unit to deny access to the services or actions" does it at source

upvoted 2 times

Ruffyit 1 year, 6 months ago

D. Service control policies (SCPs) are one type of policy that you can use to manage your organization. SCPs offer central control over the maximum available permissions for all accounts in your organization, allowing you to ensure your accounts stay within your organization's access control guidelines. See

https://docs.aws.amazon.com/organizations/latest/userguide/orgs_manage_policies_scp.html.

upvoted 2 times

mach2022 1 year, 7 months ago

is D because of Deeznuts

upvoted 2 times

the_bong_lord 1 year, 4 months ago

gottem

upvoted 1 times

xplusfb 1 year, 7 months ago

Selected Answer: D

Its very clear question answer is D

upvoted 3 times

TariqKipkemei 1 year, 9 months ago

Selected Answer: D

Service control policies (SCPs) are a type of organization policy that you can use to manage permissions in your organization. SCPs offer central control over the maximum available permissions for all accounts in your organization. SCPs help you to ensure your accounts stay within your organization's access control guidelines.

upvoted 2 times

Guru4Cloud 1 year, 10 months ago

Selected Answer: D

D. Service control policies (SCPs) are one type of policy that you can use to manage your organization. SCPs offer central control over the maximum available permissions for all accounts in your organization, allowing you to ensure your accounts stay within your organization's access control guidelines. See

https://docs.aws.amazon.com/organizations/latest/userguide/orgs_manage_policies_scp.html.

upvoted 2 times

Bmarodi 2 years ago

Selected Answer: D

I vote for option D by Creating a service control policy (SCP) in the root organizational unit to deny access to the services or actions, meets the requirements.

upvoted 2 times

Buruguduystunstugudunstuy 2 years, 5 months ago

Selected Answer: D

To limit access to specific services or actions in all of the team's AWS accounts and maintain a single point where permissions can be managed, the solutions architect should create a service control policy (SCP) in the root organizational unit to deny access to the services or actions (Option D).

Service control policies (SCPs) are policies that you can use to set fine-grained permissions for your AWS accounts within your organization. SCPs are attached to the root of the organizational unit (OU) or to individual accounts, and they specify the permissions that are allowed or denied for the accounts within the scope of the policy. By creating an SCP in the root organizational unit, the security team can set permissions for all of the accounts in the organization from a single location, ensuring that the permissions are consistently applied across all accounts.

upvoted 6 times

career360guru 2 years, 5 months ago

Selected Answer: D

Option D

upvoted 1 times

Wpcorgan 2 years, 6 months ago

D is incorrect

upvoted 1 times

babaxoxo 2 years, 6 months ago

an organization and requires single point place to manage permissions

upvoted 2 times

goatbernard 2 years, 7 months ago

Selected Answer: D

SCP for organization

upvoted 4 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #169

Topic 1

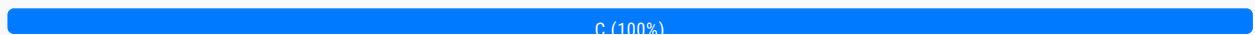
A company is concerned about the security of its public web application due to recent web attacks. The application uses an Application Load Balancer (ALB). A solutions architect must reduce the risk of DDoS attacks against the application.

What should the solutions architect do to meet this requirement?

- A. Add an Amazon Inspector agent to the ALB.
- B. Configure Amazon Macie to prevent attacks.
- C. Enable AWS Shield Advanced to prevent attacks. **Most Voted**
- D. Configure Amazon GuardDuty to monitor the ALB.

Correct Answer: C

Community vote distribution



Comments

studynoplay **Highly Voted** 2 years, 1 month ago

What's going on, suddenly the questions are so easy
upvoted 12 times

Sutariya 1 year, 10 months ago

Its due to confidence level going up after experience.
upvoted 8 times

awashenko **Highly Voted** 1 year, 8 months ago

Selected Answer: C

When you see DDOS immediately think Shield
upvoted 7 times

satyaamm **Most Recent** 4 months ago

Selected Answer: C

AWS Shield is most suitable for DDoS attacks.
upvoted 1 times

upvoted 1 times

PaulGa 7 months, 2 weeks ago

Selected Answer: C

Ans C - DDoS under AWS = Shield/ Shield Advanced

upvoted 2 times

Ruffyt 1 year, 6 months ago

When you see DDOS immediately think Shield

upvoted 6 times

TariqKipkemei 1 year, 9 months ago

Selected Answer: C

AWS Shield is a managed DDoS protection service that safeguards applications running on AWS.

upvoted 2 times

Guru4Cloud 1 year, 10 months ago

Selected Answer: C

Enable AWS Shield Advanced to prevent attacks.

upvoted 2 times

cookieMr 1 year, 11 months ago

Selected Answer: C

By enabling Shield Advanced, the web application benefits from automatic protection against common and sophisticated DDoS attacks. It utilizes advanced detection and mitigation techniques, including ML algorithms and traffic analysis, to provide effective DDoS protection.

It also includes features like real-time monitoring, attack notifications, and detailed attack reports.

A is not related to DDoS protection. Amazon Inspector is a security assessment service that helps identify vulnerabilities and security issues in applications and EC2.

B is also not the appropriate solution. Macie is a service that uses machine learning to discover, classify, and protect sensitive data stored in AWS. It focuses on data security and protection, not specifically on DDoS prevention.

D is not the most effective solution. GuardDuty is a threat detection service that analyzes events and network traffic to identify potential security threats and anomalies. While it can provide insights into potential DDoS attacks, it does not actively prevent or mitigate them.

upvoted 4 times

techhb 2 years, 5 months ago

Explained in details here <https://medium.com/@tshemku/aws-waf-vs-firewall-manager-vs-shield-vs-shield-advanced-4c86911e94c6>

upvoted 3 times

Buruguduystunstugudunstuy 2 years, 5 months ago

Selected Answer: C

To reduce the risk of DDoS attacks against the application, the solutions architect should enable AWS Shield Advanced (Option C).

AWS Shield is a managed Distributed Denial of Service (DDoS) protection service that helps protect web applications running on AWS from DDoS attacks. AWS Shield Advanced is an additional layer of protection that provides enhanced DDoS protection capabilities, including proactive monitoring and automatic inline mitigations, to help protect against even the largest and most sophisticated DDoS attacks. By enabling AWS Shield Advanced, the solutions architect can help protect the application from DDoS attacks and reduce the risk of disruption to the application.

upvoted 5 times

career360guru 2 years, 5 months ago

Selected Answer: C

C is right answer

upvoted 2 times

Wpcorgan 2 years, 6 months ago

C is correct

upvoted 2 times

goatbernard 2 years, 7 months ago

Selected Answer: C

AWS Shield Advanced

upvoted 4 times

Nigma 2 years, 7 months ago

DDOS = AWS Shield

upvoted 5 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #170

Topic 1

A company's web application is running on Amazon EC2 instances behind an Application Load Balancer. The company recently changed its policy, which now requires the application to be accessed from one specific country only.

Which configuration will meet this requirement?

- A. Configure the security group for the EC2 instances.
- B. Configure the security group on the Application Load Balancer.
- C. Configure AWS WAF on the Application Load Balancer in a VPC. **Most Voted**
- D. Configure the network ACL for the subnet that contains the EC2 instances.

Correct Answer: C

Community vote distribution

C (100%)

Comments

handyplazt **Highly Voted** 2 years, 6 months ago

Selected Answer: C

Geographic (Geo) Match Conditions in AWS WAF. This new condition type allows you to use AWS WAF to restrict application access based on the geographic location of your viewers. With geo match conditions you can choose the countries from which AWS WAF should allow access.

<https://aws.amazon.com/about-aws/whats-new/2017/10/aws-waf-now-supports-geographic-match/>

upvoted 29 times

Buruguduystunstugudunstuy **Highly Voted** 2 years, 5 months ago

Selected Answer: C

To meet the requirement of allowing the web application to be accessed from one specific country only, the company should configure AWS WAF (Web Application Firewall) on the Application Load Balancer in a VPC (Option C).

AWS WAF is a web application firewall service that helps protect web applications from common web exploits that could affect application availability, compromise security, or consume excessive resources. AWS WAF allows you to create rules that block or allow traffic based on the values of specific request parameters, such as IP address, HTTP header, or query string value. By configuring AWS WAF on the Application Load Balancer and creating rules that allow traffic from a specific country, the company can ensure that the web application is only accessible from that country.

upvoted 6 times

Dharmarajan Most Recent 4 months, 1 week ago

Selected Answer: C

AWF's Geo Match rule helps in this scenario.
upvoted 1 times

PaulGa 7 months, 2 weeks ago

Selected Answer: C

Ans C - WAF with geo-match (region or country).
<https://docs.aws.amazon.com/waf/latest/developerguide/waf-rule-statement-type-geo-match.html>
upvoted 2 times

PoolDead 10 months, 1 week ago

C --> One of the feature of WAF is Access Control: Implement IP whitelisting and blacklisting to allow or block traffic from specific IP addresses or address ranges. This can be useful for restricting access to your web application to trusted users or regions.
upvoted 3 times

Ruffyt 1 year, 6 months ago

Geographic (Geo) Match Conditions in AWS WAF. This new condition type allows you to use AWS WAF to restrict application access based on the geographic location of your viewers. With geo match conditions you can choose the countries from which AWS WAF should allow access.
<https://aws.amazon.com/about-aws/whats-new/2017/10/aws-waf-now-supports-geographic-match/>
upvoted 4 times

Guru4Cloud 1 year, 10 months ago

Selected Answer: C

C. Configure AWS WAF on the Application Load Balancer in a VPC
upvoted 1 times

Sutariya 1 year, 10 months ago

We can use AWS WAF to configure access control rule to access from specific location.
upvoted 1 times

cookieMr 1 year, 11 months ago

Selected Answer: C

By configuring AWS WAF on the ALB in a VPC, you can apply access control rules based on the geographic location of the incoming requests. AWS WAF allows you to create rules that include conditions based on the IP addresses' country of origin. You can specify the desired country and deny access to requests originating from any other country by leveraging AWS WAF's Geo Match feature.

Option A and option B focus on network-level access control and do not provide country-specific filtering capabilities.

Option D is not the ideal solution for restricting access based on country. Network ACLs primarily control traffic at the subnet level based on IP addresses and port numbers, but they do not have built-in capabilities for country-based filtering.
upvoted 5 times

Abrar2022 2 years ago

Configure AWS WAF for Geo Match Policy
upvoted 3 times

aba2s 2 years, 5 months ago

Selected Answer: C

Source from an AWS link
Geographic (Geo) Match Conditions in AWS WAF. This condition type allows you to use AWS WAF to restrict application access based on the geographic location of your viewers.
With geo match conditions you can choose the countries from which AWS WAF should allow access.
upvoted 3 times

techhb 2 years, 5 months ago

Selected Answer: C

Selected Answer: C

WAF Shield Advanced for DDOS,
GuardDuty is a continuous monitoring service that alerts you of potential threats, while Inspector is a one-time assessment service that provides a report of vulnerabilities and deviations from best practices.
upvoted 2 times

career360guru 2 years, 5 months ago

Selected Answer: C

OptionC. Configure WAF for Geo Match Policy
upvoted 2 times

Wpcorgan 2 years, 6 months ago

C is correct
upvoted 2 times

mricee9 2 years, 6 months ago

Selected Answer: C

C
<https://aws.amazon.com/about-aws/whats-new/2017/10/aws-waf-now-supports-geographic-match/>
upvoted 3 times

Nigma 2 years, 7 months ago

C. WAF with ALB is the right option
upvoted 2 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #171

Topic 1

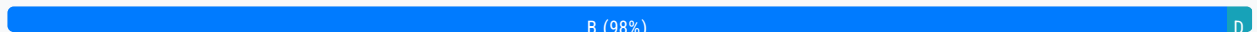
A company provides an API to its users that automates inquiries for tax computations based on item prices. The company experiences a larger number of inquiries during the holiday season only that cause slower response times. A solutions architect needs to design a solution that is scalable and elastic.

What should the solutions architect do to accomplish this?

- A. Provide an API hosted on an Amazon EC2 instance. The EC2 instance performs the required computations when the API request is made.
- B. Design a REST API using Amazon API Gateway that accepts the item names. API Gateway passes item names to AWS Lambda for tax computations. **Most Voted**
- C. Create an Application Load Balancer that has two Amazon EC2 instances behind it. The EC2 instances will compute the tax on the received item names.
- D. Design a REST API using Amazon API Gateway that connects with an API hosted on an Amazon EC2 instance. API Gateway accepts and passes the item names to the EC2 instance for tax computations.

Correct Answer: B

Community vote distribution



Comments

bullrem **Highly Voted** 2 years, 4 months ago

Selected Answer: B

Option D is similar to option B in that it uses Amazon API Gateway to handle the API requests, but it also includes an EC2 instance to perform the tax computations. However, using an EC2 instance in this way is less scalable and less elastic than using AWS Lambda to perform the computations. An EC2 instance is a fixed resource and requires manual scaling and management, while Lambda is an event-driven, serverless compute service that automatically scales with the number of requests, making it more suitable for handling variable workloads and reducing response times during high traffic periods. Additionally, Lambda is more cost-efficient than EC2 instances, as you only pay for the compute time consumed by your functions, making it a more cost-effective solution.

upvoted 29 times

career360guru **Highly Voted** 2 years, 5 months ago

Selected Answer: B

Option B. Though D is also possible B is more scalable as Lambda will autoscale to meet the dynamic load.

upvoted 6 times

pentium75 1 year, 5 months ago

Even though you can scale EC2 instances (which is less flexible as Lambda), D does not mention scaling but rather a single EC2 instance. Thus it's clearly out.

upvoted 3 times

satyaamm Most Recent 4 months ago

Selected Answer: B

Both API Gateway and Lambda are suitable as they are scalable.

upvoted 1 times

PaulGa 7 months, 2 weeks ago

Selected Answer: B

Ans B - I was going for option D... but then I saw bullrem's response (1 year, 9 mth ago):

"Option D is similar to option B in that it uses Amazon API Gateway to handle the API requests, but it also includes an EC2 instance to perform the tax computations. However, using an EC2 instance in this way is less scalable and less elastic than using AWS Lambda to perform the computations."

upvoted 2 times

lofzee 1 year ago

Selected Answer: B

b. easy

upvoted 2 times

TruthWS 1 year, 2 months ago

Option B: Lamda = scalable and elastic

upvoted 3 times

Uzbekistan 1 year, 2 months ago

Selected Answer: B

Option B leverages AWS Lambda, which is a serverless compute service that automatically scales in response to incoming requests. When a request is made to the API hosted on Amazon API Gateway, API Gateway triggers the associated AWS Lambda function, passing the item names as input parameters. The Lambda function then performs the tax computations based on the provided item names. AWS Lambda automatically manages the compute capacity, ensuring that there is no need to provision or manage servers. This serverless architecture offers scalability and elasticity, as Lambda functions can scale out to handle a larger number of inquiries during the holiday season and scale in during periods of lower demand. Additionally, AWS Lambda is a fully managed service, reducing operational overhead for the company.

upvoted 2 times

app12 1 year, 4 months ago

The thing that bothers me about B is that the request sends the name and then based on the name, the tax is calculated. How do you calculate a value e.g. tax if you just have a name...

upvoted 3 times

awsgeek75 1 year, 5 months ago

Selected Answer: B

EC2 without autoscaling is not elastic so A, C & D won't be suitable. B uses AWS Lambda which is elastic and scalable by design.

upvoted 4 times

pentium75 1 year, 5 months ago

Selected Answer: B

Though EC2 can scale (even if less flexible than Lambda), neither A, C nor D involve scaling. All these answers are about a single EC2 instance or a pair of EC2 instances. The only answer that includes scaling and elasticity is B.

upvoted 5 times

LeVel 1 year, 5 months ago

LOXOL 1 year, 5 months ago

I agree.
upvoted 1 times

Ruffyit 1 year, 6 months ago

scalable and elastic = serverless = API gateway and AWS Lambda
upvoted 3 times

paniya93 1 year, 8 months ago

Selected Answer: B

in 002 answer is B. Why is that?
upvoted 1 times

vijaykamal 1 year, 8 months ago

Selected Answer: B

Options A, C, and D involve EC2 instances, which are not as inherently scalable and elastic as serverless AWS Lambda functions, and they would require more manual management and operational overhead. Therefore, option B is the most appropriate choice for a scalable and elastic API solution.
upvoted 3 times

Guru4Cloud 1 year, 9 months ago

Selected Answer: B

REST API using Amazon API Gateway and integrating it with AWS Lambda (option B) is the recommended approach to achieve a scalable and elastic solution for the company's API during the holiday season.

No good EC2 in this case
using an EC2 instance in this way is less scalable and less elastic than using AWS Lambda to perform the computations
upvoted 3 times

TariqKipkemei 1 year, 9 months ago

Selected Answer: B

scalable and elastic = serverless = API gateway and AWS Lambda
upvoted 2 times

Guru4Cloud 1 year, 10 months ago

B) Design a REST API using Amazon API Gateway that accepts the item names. API Gateway passes item names to AWS Lambda for tax computations.

This option provides the most scalable and elastic solution:

API Gateway handles creating the REST API frontend to receive requests
Lambda functions scale automatically to handle spikes in traffic during peak seasons
No servers to manage for the computations, providing high scalability
upvoted 3 times

cookieMr 1 year, 11 months ago

Selected Answer: B

Option A (hosting an API on an Amazon EC2 instance) would require manual management and scaling of the EC2 instances, making it less scalable and elastic compared to a serverless solution.

Option C (creating an Application Load Balancer with EC2 instances for tax computations) also involves manual management of the instances and does not offer the same level of scalability and elasticity as a serverless solution.

Option D (designing a REST API using API Gateway and connecting it with an API hosted on an EC2 instance) adds unnecessary complexity and management overhead. It is more efficient to directly integrate API Gateway with AWS Lambda for tax computations.

Therefore, designing a REST API using Amazon API Gateway and integrating it with AWS Lambda (option B) is the recommended approach to achieve a scalable and elastic solution for the company's API during the holiday season.
upvoted 3 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #172

Topic 1

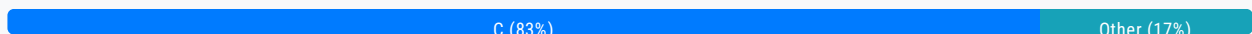
A solutions architect is creating a new Amazon CloudFront distribution for an application. Some of the information submitted by users is sensitive. The application uses HTTPS but needs another layer of security. The sensitive information should be protected throughout the entire application stack, and access to the information should be restricted to certain applications.

Which action should the solutions architect take?

- A. Configure a CloudFront signed URL.
- B. Configure a CloudFront signed cookie.
- C. Configure a CloudFront field-level encryption profile. **Most Voted**
- D. Configure CloudFront and set the Origin Protocol Policy setting to HTTPS Only for the Viewer Protocol Policy.

Correct Answer: C

Community vote distribution



Comments

Bobbybash **Highly Voted** 2 years, 6 months ago

CCCCCCCC

Field-level encryption allows you to enable your users to securely upload sensitive information to your web servers. The sensitive information provided by your users is encrypted at the edge, close to the user, and remains encrypted throughout your entire application stack. This encryption ensures that only applications that need the data—and have the credentials to decrypt it—are able to do so.

upvoted 58 times

jatric **Most Recent** 11 months, 1 week ago

Selected Answer: C

field level encryption allow to protect sensitive information throughout the application stack

upvoted 2 times

zinabu 1 year, 2 months ago

With Amazon CloudFront, you can enforce secure end-to-end connections to origin servers by using HTTPS. Field-level encryption adds an additional layer of security that lets you protect specific data throughout system processing so that only certain applications can see it.

Field-level encryption allows you to enable your users to securely upload sensitive information to your web servers. The sensitive information provided by your users is encrypted at the edge, close to the user, and remains encrypted throughout your entire application stack. This encryption ensures that only applications that need the data—and have the credentials to decrypt it—are able to do so.

upvoted 4 times

huzafaharoun 1 year, 3 months ago

C:

<https://docs.aws.amazon.com/AmazonCloudFront/latest/DeveloperGuide/field-level-encryption.html>

upvoted 3 times

NayeraB 1 year, 3 months ago

Selected Answer: C

C is the only one that addresses handling sensitive information.

upvoted 3 times

bujuman 1 year, 4 months ago

Selected Answer: C

Reviewing my first vote after research. It seems that C is the best answer:

<https://docs.aws.amazon.com/AmazonCloudFront/latest/DeveloperGuide/field-level-encryption.html>

upvoted 2 times

awsgeek75 1 year, 5 months ago

Selected Answer: C

A if for fetch. B requires cookies. D just enforces HTTPS which is already mentioned for the solution (CloudFront only allows HTTPS) and does not add another layer of security.

C provides field level encryption security which is another layer of security.

upvoted 2 times

master9 1 year, 6 months ago

Selected Answer: A

Please go through below link:

<https://docs.aws.amazon.com/AmazonCloudFront/latest/DeveloperGuide/private-content-signed-urls.html>

upvoted 2 times

pentium75 1 year, 5 months ago

This is about controlling access for downloads (making sure that the download request is coming from an authenticated user), it has nothing to do with protecting data that is sent to the application.

upvoted 3 times

Leo1688 1 year, 6 months ago

cccc,this link <https://docs.aws.amazon.com/AmazonCloudFront/latest/DeveloperGuide/field-level-encryption.html>

upvoted 3 times

vijaykamal 1 year, 8 months ago

Selected Answer: C

Options A and B (signed URL and signed cookie) are used for controlling access to specific resources and are typically used for restricting access based on URLs or cookies. They do not provide field-level encryption for sensitive data within HTTP requests.

Option D (configuring CloudFront with the Origin Protocol Policy set to HTTPS Only for the Viewer Protocol Policy) is related to enforcing HTTPS communication between CloudFront and the viewer (end-user). While important for security, it doesn't address the specific requirement of protecting sensitive data within the application stack.

upvoted 4 times

Guru4Cloud 1 year, 10 months ago

Selected Answer: C

C) Configure a CloudFront field-level encryption profile.

Field-level encryption allows you to encrypt sensitive information at the edge before distributing content through CloudFront. It provides an additional layer of security for sensitive user-submitted data.

The other options would not provide field-level encryption
upvoted 2 times

mr_D3v1n3 1 year, 10 months ago

Would the HTTPS imply that the cert was signed by a CA
upvoted 1 times

cookieMr 1 year, 11 months ago

Selected Answer: C

Option A and Option B are used for controlling access to specific resources or content based on signed URLs or cookies. While they provide security and access control, they do not provide field-level encryption for sensitive data within the requests.

Option D ensures that communication between the viewer and CloudFront is encrypted with HTTPS. However, it does not specifically address the protection and encryption of sensitive information within the application stack.

Therefore, the most appropriate action to protect sensitive information throughout the entire application stack and restrict access to certain applications is to configure a CloudFront field-level encryption profile (Option C).

upvoted 3 times

Jeeva28 2 years ago

Selected Answer: C

With Amazon CloudFront, you can enforce secure end-to-end connections to origin servers by using HTTPS. Field-level encryption adds an additional layer of security that lets you protect specific data throughout system processing so that only certain applications can see it.

upvoted 2 times

WheracanIstart 2 years, 3 months ago

Selected Answer: C

<https://docs.aws.amazon.com/AmazonCloudFront/latest/DeveloperGuide/field-level-encryption.html>

"Field-level encryption allows you to enable your users to securely upload sensitive information to your web servers. The sensitive information provided by your users is encrypted at the edge, close to the user, and remains encrypted throughout your entire application stack".

upvoted 4 times

bdp123 2 years, 4 months ago

Selected Answer: C

<https://docs.aws.amazon.com/AmazonCloudFront/latest/DeveloperGuide/field-level-encryption.html>

"With Amazon CloudFront, you can enforce secure end-to-end connections to origin servers by using HTTPS. Field-level encryption adds an additional layer of security that lets you protect specific data throughout system processing so that only certain applications can see it."

upvoted 4 times

ProfXsamson 2 years, 4 months ago

C, field-level encryption should be used when necessary to protect sensitive data.

upvoted 2 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #173

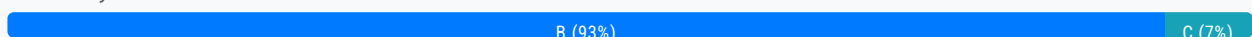
Topic 1

A gaming company hosts a browser-based application on AWS. The users of the application consume a large number of videos and images that are stored in Amazon S3. This content is the same for all users.

The application has increased in popularity, and millions of users worldwide accessing these media files. The company wants to provide the files to the users while reducing the load on the origin.

Which solution meets these requirements MOST cost-effectively?

- A. Deploy an AWS Global Accelerator accelerator in front of the web servers.
- B. Deploy an Amazon CloudFront web distribution in front of the S3 bucket. **Most Voted**
- C. Deploy an Amazon ElastiCache for Redis instance in front of the web servers.
- D. Deploy an Amazon ElastiCache for Memcached instance in front of the web servers.

Correct Answer: B*Community vote distribution***Comments****Nigma** **Highly Voted** 2 years ago

B. Cloud front is best for content delivery. Global Accelerator is best for non-HTTP (TCP/UDP) cases and supports HTTP cases as well but with static IP (elastic IP) or anycast IP address only.

upvoted 23 times

cookieMr **Highly Voted** 1 year, 5 months ago**Selected Answer: B**

Option A is not the most cost-effective solution for this scenario. While Global Accelerator can improve global application performance, it is primarily used for accelerating TCP and UDP traffic, such as gaming and real-time applications, rather than serving static media files.

Options C and D are used for caching frequently accessed data in-memory to improve application performance. However, they are not specifically designed for caching and serving media files like CloudFront, and therefore, may not provide the same cost-effectiveness and scalability for this use case.

Hence, deploying an CloudFront web distribution in front of the S3 is the most cost-effective solution for delivering media files

hence, deploying an CloudFront web distribution in front of the S3 is the most cost-effective solution for delivering media files to millions of users worldwide while reducing the load on the origin.

upvoted 6 times

rlamberti Most Recent 1 year, 1 month ago

Selected Answer: B

CloudFront will cache the data in Edge Locations, offloading it partially from the source location (s3)

B looks good to me.

upvoted 3 times

TariqKipkemei 1 year, 3 months ago

Selected Answer: B

Deploy an Amazon CloudFront web distribution in front of the S3 bucket

upvoted 2 times

Guru4Cloud 1 year, 3 months ago

Selected Answer: B

B) Deploy an Amazon CloudFront web distribution in front of the S3 bucket.

CloudFront is the most cost-effective solution for this use case because:

CloudFront can cache static assets like videos and images at edge locations closer to users. This improves performance.

Serving files from the CloudFront cache reduces load on the S3 origin.

CloudFront pricing is very low for data transfer and requests.

upvoted 2 times

Kiki_Pass 1 year, 4 months ago

Selected Answer: B

ElasticCache is for DB Cache(RDS) nor for S3

upvoted 1 times

kruasan 1 year, 7 months ago

Selected Answer: B

ElastiCache, enhances the performance of web applications by quickly retrieving information from fully-managed in-memory data stores. It utilizes Memcached and Redis, and manages to considerably reduce the time your applications would, otherwise, take to read data from disk-based databases.

Amazon CloudFront supports dynamic content from HTTP and WebSocket protocols, which are based on the Transmission Control Protocol (TCP) protocol. Common use cases include dynamic API calls, web pages and web applications, as well as an application's static files such as audio and images. It also supports on-demand media streaming over HTTP.

AWS Global Accelerator supports both User Datagram Protocol (UDP) and TCP-based protocols. It is commonly used for non-HTTP use cases, such as gaming, IoT and voice over IP. It is also good for HTTP use cases that need static IP addresses or fast regional failover

upvoted 4 times

LuckyAro 1 year, 11 months ago

Selected Answer: C

The company wants to provide the files to the users while reducing the load on the origin.

Cloudfront speeds-up content delivery but I'm not sure it reduces the load on the origin.

Some form of caching would cache content and deliver to users without going to the origin for each request.

upvoted 2 times

pentium75 11 months, 3 weeks ago

ElastiCache for Redis (C) can be used by an application to store key-value pairs. It does not cache videos or images and it is not an automatic process (the application can put and retrieve values).

upvoted 2 times

pentium75 11 months, 3 weeks ago

"Some form of caching would cache content and deliver to users without going to the origin for each request", isn't this EXACTLY what CloudFront does?

upvoted 4 times

awsgeek75 11 months ago

CloudFront's only main purpose is to serve these kind of scenarios!

upvoted 3 times

Buruguduystunstugudunstuy 1 year, 11 months ago

Selected Answer: B

To provide media files to users while reducing the load on the origin and meeting the requirements cost-effectively, the gaming company should deploy an Amazon CloudFront web distribution in front of the S3 bucket (Option B).

CloudFront is a content delivery network (CDN) that speeds up the delivery of static and dynamic web content, such as images and videos, to users. By using CloudFront, the media files will be served to users from the edge location that is closest to them, resulting in faster delivery and a better user experience. CloudFront can also handle the high traffic and large number of requests expected from the millions of users, ensuring that the media files are available and accessible to users around the world.

upvoted 4 times

techhb 1 year, 11 months ago

Please dont post ChatGPT answers here, chatgpt keeps on changing its answers, its not the right way to copy paste, thanks.

upvoted 2 times

Bofi 1 year, 9 months ago

why not? if the answers are correct and offer best possible explanation for the wrong options, I see no reason why it shouldn't be posted here. Also, most of his answers were right, although reasons for the wrong options were sometimes lacking, but all in all, his responses were very good.

upvoted 2 times

ocbn3wby 1 year, 10 months ago

Answers are mostly correct. Only a small percentage were wrong

upvoted 2 times

ocbn3wby 1 year, 10 months ago

Woaaaa! I always wondered where this kind of logic and explanation came from in this guy's answers. Nice catch TECHHB!

upvoted 3 times

career360guru 1 year, 12 months ago

Selected Answer: B

Option B

upvoted 1 times

k1kavi1 1 year, 12 months ago

Selected Answer: B

Agreed

upvoted 1 times

rewdboy 2 years ago

Selected Answer: B

B is the correct answer

upvoted 1 times

Wpcorgan 2 years ago

B is correct

upvoted 2 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #174

Topic 1

A company has a multi-tier application that runs six front-end web servers in an Amazon EC2 Auto Scaling group in a single Availability Zone behind an Application Load Balancer (ALB). A solutions architect needs to modify the infrastructure to be highly available without modifying the application.

Which architecture should the solutions architect choose that provides high availability?

- A. Create an Auto Scaling group that uses three instances across each of two Regions.
- B. Modify the Auto Scaling group to use three instances across each of two Availability Zones. **Most Voted**
- C. Create an Auto Scaling template that can be used to quickly create more instances in another Region.
- D. Change the ALB in front of the Amazon EC2 instances in a round-robin configuration to balance traffic to the web tier.

Correct Answer: B

Community vote distribution

B (100%)

Comments

Nigma **Highly Voted** 1 year, 7 months ago

B. auto scaling groups can not span multi region
upvoted 30 times

cookieMr **Highly Voted** 11 months, 3 weeks ago

Selected Answer: B

Option A (creating an Auto Scaling group across two Regions) introduces additional complexity and potential replication challenges, which may not be necessary for achieving high availability within a single Region.

Option C (creating an Auto Scaling template for another Region) suggests multi-region redundancy, which may not be the most straightforward solution for achieving high availability without modifying the application.

Option D (changing the ALB to a round-robin configuration) does not provide the desired high availability. Round-robin configuration alone does not ensure fault tolerance and does not leverage multiple Availability Zones for resilience.

Hence, modifying the Auto Scaling group to use three instances across each of two Availability Zones is the appropriate choice to provide high availability for the multi-tier application.

upvoted 10 times

satyaamm Most Recent 4 months ago

Selected Answer: B

Modifying the ALB is the most suitable option here.
upvoted 1 times

TariqKipkemei 9 months ago

Selected Answer: B

Modify the Auto Scaling group to use three instances across each of two Availability Zones
upvoted 4 times

Guru4Cloud 10 months ago

Selected Answer: B

Option B. Modify the Auto Scaling group to use three instances across each of the two Availability Zones.
upvoted 3 times

techhb 1 year, 5 months ago

B. auto scaling groups cannot span multi region
upvoted 6 times

Buruguduystunstugudunstuy 1 year, 5 months ago

Selected Answer: B

Option B. Modify the Auto Scaling group to use three instances across each of the two Availability Zones.

This option would provide high availability by distributing the front-end web servers across multiple Availability Zones. If there is an issue with one Availability Zone, the other Availability Zone would still be available to serve traffic. This would ensure that the application remains available and highly available even if there is a failure in one of the Availability Zones.
upvoted 5 times

career360guru 1 year, 5 months ago

Selected Answer: B

Option B
upvoted 1 times

k1kavi1 1 year, 6 months ago

Selected Answer: B

Agreed
upvoted 1 times

Shasha1 1 year, 6 months ago

B
option B This architecture provides high availability by having multiple Availability Zones hosting the same application. This allows for redundancy in case one Availability Zone experiences downtime, as traffic can be served by the other Availability Zone. This solution also increases scalability and performance by allowing traffic to be spread across two Availability Zones.
upvoted 2 times

mricee9 1 year, 6 months ago

Selected Answer: B

B is right
upvoted 2 times

Wpcorgan 1 year, 6 months ago

B is correct
upvoted 1 times

xua81376 1 year, 6 months ago

B auto scaling i multiple AZ
upvoted 1 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #175

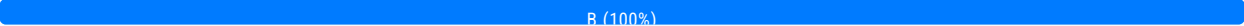
Topic 1

An ecommerce company has an order-processing application that uses Amazon API Gateway and an AWS Lambda function. The application stores data in an Amazon Aurora PostgreSQL database. During a recent sales event, a sudden surge in customer orders occurred. Some customers experienced timeouts, and the application did not process the orders of those customers.

A solutions architect determined that the CPU utilization and memory utilization were high on the database because of a large number of open connections. The solutions architect needs to prevent the timeout errors while making the least possible changes to the application.

Which solution will meet these requirements?

- A. Configure provisioned concurrency for the Lambda function. Modify the database to be a global database in multiple AWS Regions.
- B. Use Amazon RDS Proxy to create a proxy for the database. Modify the Lambda function to use the RDS Proxy endpoint instead of the database endpoint. **Most Voted**
- C. Create a read replica for the database in a different AWS Region. Use query string parameters in API Gateway to route traffic to the read replica.
- D. Migrate the data from Aurora PostgreSQL to Amazon DynamoDB by using AWS Database Migration Service (AWS DMS). Modify the Lambda function to use the DynamoDB table.

Correct Answer: B*Community vote distribution*B (100%)**Comments****handyplazt** **Highly Voted** 2 years, 6 months ago**Selected Answer: B**

Many applications, including those built on modern serverless architectures, can have a large number of open connections to the database server and may open and close database connections at a high rate, exhausting database memory and compute resources. Amazon RDS Proxy allows applications to pool and share connections established with the database, improving database efficiency and application scalability.

<https://aws.amazon.com/id/rds/proxy/>

upvoted 38 times

rexxxx_x 1 year, 4 months ago

Are you sure?
upvoted 2 times

cookieMr **Highly Voted** 1 year, 11 months ago

Selected Answer: B

Option A (configuring provisioned concurrency and creating a global database) does not directly address the high connection utilization issue on the database, and creating a global database may introduce additional complexity without immediate benefit to solving the timeout errors.

Option C (creating a read replica in a different AWS Region) introduces additional data replication and management complexity, which may not be necessary to address the timeout errors.

Option D (migrating to Amazon DynamoDB) involves a significant change in the data storage technology and requires modifying the application to use DynamoDB instead of Aurora PostgreSQL. This may not be the most suitable solution when the goal is to make minimal changes to the application.

Therefore, using Amazon RDS Proxy and modifying the Lambda function to use the RDS Proxy endpoint is the recommended solution to prevent timeout errors and reduce the impact on the database during peak loads.

upvoted 11 times

satyaamm **Most Recent** 4 months ago

Selected Answer: B

RDS proxy helps tackle the problem of sudden hikes as it plays a medium between the users and the application

upvoted 1 times

lofzee 1 year ago

Selected Answer: B

Amazon RDS Proxy is a fully managed, highly available database proxy for Amazon Relational Database Service (RDS) that makes applications more scalable, more resilient to database failures, and more secure.

B

upvoted 3 times

awsgeek75 1 year, 4 months ago

Selected Answer: B

Connection problems causing high CPU and Memory usage? Use RDS proxy!

upvoted 3 times

tonybuivannghia 8 months, 3 weeks ago

Improving database efficiency by reducing the stress on database resources (e.g., CPU, RAM) and minimize open connections (and timeouts).

upvoted 2 times

TariqKipkemei 1 year, 9 months ago

Selected Answer: B

Use Amazon RDS Proxy to create a proxy for the database. Modify the Lambda function to use the RDS Proxy endpoint instead of the database endpoint.

upvoted 3 times

Guru4Cloud 1 year, 10 months ago

Selected Answer: B

using Amazon RDS Proxy and modifying the Lambda function to use the RDS Proxy endpoint is the recommended solution to prevent timeout errors and reduce the impact on the database during peak loads.

upvoted 2 times

sairam 2 years, 4 months ago

I also think the answer is B. However can RDS Proxy be used with Amazon Aurora PostgreSQL database?

upvoted 2 times

everfly 2 years, 3 months ago

RDS Proxy can be used with Aurora

<https://docs.aws.amazon.com/AmazonRDS/latest/AuroraUserGuide/rds-proxy.html>

upvoted 6 times

gustavtd 2 years, 5 months ago

Selected Answer: B

I expect a answer with database replica but there is not, so B is most suitable

upvoted 2 times

Buruguduystunstugudunstuy 2 years, 5 months ago

Selected Answer: B

Option B. Use Amazon RDS Proxy to create a proxy for the database. Modify the Lambda function to use the RDS Proxy endpoint instead of the database endpoint.

Using Amazon RDS Proxy can help reduce the number of connections to the database and improve the performance of the application. RDS Proxy establishes a connection pool to the database and routes connections to the available connections in the pool. This can help reduce the number of open connections to the database and improve the performance of the application. The Lambda function can be modified to use the RDS Proxy endpoint instead of the database endpoint to take advantage of this improvement.

upvoted 2 times

Buruguduystunstugudunstuy 2 years, 5 months ago

Option A is not a valid solution because configuring provisioned concurrency for the Lambda function does not address the issue of high CPU utilization and memory utilization on the database.

Option C is not a valid solution because creating a read replica in a different Region does not address the issue of high CPU utilization and memory utilization on the database.

Option D is not a valid solution because migrating the data from Aurora PostgreSQL to DynamoDB would require significant changes to the application and may not be the best solution for this particular problem.

upvoted 3 times

BENICE 2 years, 5 months ago

Option --- B

upvoted 2 times

career360guru 2 years, 5 months ago

Selected Answer: B

As it is mentioned that issue was due to high CPU and Memory due to many open connections to DB, B is the right answer.

upvoted 2 times

Shasha1 2 years, 6 months ago

B

Using Amazon RDS Proxy will allow the application to handle more connections and higher loads without timeouts, while making the least possible changes to the application. The RDS Proxy will enable connection pooling, allowing multiple connections from the Lambda function to be served from a single proxy connection. This will reduce the number of open connections on the database, which is causing high CPU and memory utilization

upvoted 4 times

Wpcorgan 2 years, 6 months ago

B is correct

upvoted 2 times

xua81376 2 years, 6 months ago

B - Proxy to manage connections

upvoted 3 times

hahaxoxo 2 years, 6 months ago

guyana 2 years, 6 months ago

Selected Answer: B

Issue related to opening many connections and the solution requires least code changes so B satisfies the conditions
upvoted 8 times

Nigma 2 years, 7 months ago

Correct B
upvoted 2 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #176

Topic 1

An application runs on Amazon EC2 instances in private subnets. The application needs to access an Amazon DynamoDB table.

What is the MOST secure way to access the table while ensuring that the traffic does not leave the AWS network?

- A. Use a VPC endpoint for DynamoDB. **Most Voted**
- B. Use a NAT gateway in a public subnet.
- C. Use a NAT instance in a private subnet.
- D. Use the internet gateway attached to the VPC.

Correct Answer: A*Community vote distribution*

A (100%)

Comments**mabotega** **Highly Voted** 1 year, 7 months ago**Selected Answer: A**

VPC endpoints for service in private subnets

<https://docs.aws.amazon.com/amazondynamodb/latest/developerguide/vpc-endpoints-dynamodb.html>

upvoted 18 times

cookieMr **Highly Voted** 11 months, 3 weeks ago

Option B (using a NAT gateway in a public subnet) and option C (using a NAT instance in a private subnet) are not the most secure options because they involve routing traffic through a network address translation (NAT) device, which requires an internet gateway and traverses the public internet.

Option D (using the internet gateway attached to the VPC) would require routing traffic through the internet gateway, which would result in the traffic leaving the AWS network.

Therefore, the recommended and most secure approach is to use a VPC endpoint for DynamoDB to ensure private and secure access to the DynamoDB table from your EC2 instances in private subnets, without the need to traverse the internet or leave the AWS network.

upvoted 7 times

satyaamm **Most Recent** 4 months ago**Selected Answer: A**

A gateway VPC endpoint is the most suitable for accessing DynamoDB and S3.

upvoted 1 times

vijaykamal 8 months, 2 weeks ago

Selected Answer: A

Using an internet gateway (Option D) is used for enabling outbound internet connectivity from resources in your VPC. It's not the appropriate choice for securely accessing DynamoDB within your VPC.

upvoted 3 times

Ramdi1 9 months ago

Selected Answer: A

A gateway VPC Endpoint is designed for supported AWS service such as dynamo db or s3 in this case i assume the endpoint is still the valid option

upvoted 2 times

TariqKipkemei 9 months ago

Selected Answer: A

Use a VPC endpoint for DynamoDB. A VPC endpoint enables customers to privately connect to supported AWS services: Amazon DynamoDB or Amazon Simple Storage Service (Amazon S3).

upvoted 3 times

Guru4Cloud 10 months ago

Selected Answer: A

A VPC endpoint enables private connectivity between VPCs and AWS services without requiring an internet gateway, NAT device, VPN connection, or AWS Direct Connect. Traffic remains within the AWS network.

upvoted 2 times

MikeDu 10 months ago

Selected Answer: A

VPC endpoints for service in private subnets

upvoted 2 times

RashiJaiswal 11 months, 1 week ago

Selected Answer: A

VPC endpoint for dynamodb and S3

upvoted 2 times

markw92 12 months ago

VPC endpoints for DynamoDB can alleviate these challenges. A VPC endpoint for DynamoDB enables Amazon EC2 instances in your VPC to use their private IP addresses to access DynamoDB with no exposure to the public internet. Your EC2 instances do not require public IP addresses, and you don't need an internet gateway, a NAT device, or a virtual private gateway in your VPC. You use endpoint policies to control access to DynamoDB. Traffic between your VPC and the AWS service does not leave the Amazon network.

upvoted 4 times

dmt6263 1 year, 1 month ago

AAAAAAAAA

upvoted 2 times

gx2222 1 year, 2 months ago

Selected Answer: A

Option A: Use a VPC endpoint for DynamoDB - This is the correct option. A VPC endpoint for DynamoDB allows communication between resources in your VPC and Amazon DynamoDB without traversing the internet or a NAT instance, which is more secure.

upvoted 3 times

GalileoEC2 1 year, 3 months ago

A
The most secure way to access an Amazon DynamoDB table from Amazon EC2 instances in private subnets while ensuring that the traffic does not leave the AWS network is to use Amazon VPC Endpoints for DynamoDB.

Amazon VPC Endpoints enable private communication between Amazon EC2 instances in a VPC and Amazon services such as DynamoDB, without the need for an internet gateway, NAT device, or VPN connection. When you create a VPC endpoint for DynamoDB, traffic from the EC2 instances to the DynamoDB table remains within the AWS network and does not traverse the public internet.

upvoted 2 times

AIIGOD 1 year, 4 months ago

private...backend Answer A

upvoted 2 times

bdp123 1 year, 4 months ago

Selected Answer: A

<https://docs.aws.amazon.com/amazondynamodb/latest/developerguide/vpc-endpointsdynamodb.html> A VPC endpoint for DynamoDB enables Amazon EC2 instances in your VPC to use their private IP addresses to access DynamoDB with no exposure to the public internet. Your EC2 instances do not require public IP addresses, and you don't need an internet gateway, a NAT device, or a virtual private gateway in your VPC. You use endpoint policies to control access to DynamoDB. Traffic between your VPC and the AWS service does not leave the Amazon network.

upvoted 3 times

ProfXsamson 1 year, 4 months ago

ExamTopics.com should be sued for this answer tagged as Correct answer.

upvoted 4 times

mp165 1 year, 5 months ago

Selected Answer: A

A is correct. VPC end point. D exposed to the internet

upvoted 4 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #177

Topic 1

An entertainment company is using Amazon DynamoDB to store media metadata. The application is read intensive and experiencing delays. The company does not have staff to handle additional operational overhead and needs to improve the performance efficiency of DynamoDB without reconfiguring the application.

What should a solutions architect recommend to meet this requirement?

- A. Use Amazon ElastiCache for Redis.
- B. Use Amazon DynamoDB Accelerator (DAX). **Most Voted**
- C. Replicate data by using DynamoDB global tables.
- D. Use Amazon ElastiCache for Memcached with Auto Discovery enabled.

Correct Answer: B*Community vote distribution*B (100%)**Comments****techhb** **Highly Voted** 2 years, 5 months ago**Selected Answer: B**

DAX stands for DynamoDB Accelerator, and it's like a turbo boost for your DynamoDB tables. It's a fully managed, in-memory cache that speeds up the read and write performance of your DynamoDB tables, so you can get your data faster than ever before.

upvoted 25 times

cookieMr **Highly Voted** 1 year, 11 months ago**Selected Answer: B**

A. Using Amazon ElastiCache for Redis would require modifying the application code and is not specifically designed to enhance DynamoDB performance.

C. Replicating data with DynamoDB global tables would require additional configuration and operational overhead.

D. Using Amazon ElastiCache for Memcached with Auto Discovery enabled would also require application code modifications and is not specifically designed for improving DynamoDB performance.

In contrast, option B, using Amazon DynamoDB Accelerator (DAX), is the recommended solution as it is purpose-built for

enhancing DynamoDB performance without the need for application reconfiguration. DAX provides a managed caching layer that significantly reduces read latency and offloads traffic from DynamoDB tables.

upvoted 12 times

raghav2704 Most Recent 6 days ago

Selected Answer: B

elastic cache redis and memcached as mentioned in option A and D though supported by , it is not recommended with non sql database as Dynamo DB as it simply leads to more application changes with more hassle.

And replicating data with global tables will make your job even cumbersome, so dont do it.

i will go with option B

upvoted 1 times

1e22522 10 months, 2 weeks ago

Selected Answer: B

Use the turbo

upvoted 2 times

awsgeek75 1 year, 5 months ago

B: <https://aws.amazon.com/dynamodb/dax/>

improve 10x performance (marketing pitch on above link) with fully managed service so no reconfiguration or operational overhead involved.

upvoted 2 times

TariqKipkemei 1 year, 9 months ago

Selected Answer: B

Amazon DynamoDB Accelerator (DAX) is a fully managed, highly available caching service built for Amazon DynamoDB. DAX delivers up to a 10 times performance improvement—from milliseconds to microseconds—even at millions of requests per second.

[https://aws.amazon.com/dynamodb/dax/#:~:text=Amazon%20DynamoDB%20Accelerator%20\(-,DAX\),-is%20a%20fully](https://aws.amazon.com/dynamodb/dax/#:~:text=Amazon%20DynamoDB%20Accelerator%20(-,DAX),-is%20a%20fully)

upvoted 2 times

Abrar2022 2 years ago

Selected Answer: B

improve the performance efficiency of DynamoDB

upvoted 2 times

gx2222 2 years, 2 months ago

Selected Answer: B

Amazon DynamoDB Accelerator (DAX) is a fully managed, highly available, in-memory cache for DynamoDB that helps improve the read performance of DynamoDB tables. DAX provides a caching layer between the application and DynamoDB, reducing the number of read requests made directly to DynamoDB. This can significantly reduce read latencies and improve overall application performance.

upvoted 3 times

osmk 2 years, 2 months ago

B-->Applications that are read-

intensive==><https://docs.aws.amazon.com/amazondynamodb/latest/developerguide/DAX.html#DAX.use-cases>

upvoted 2 times

LuckyAro 2 years, 4 months ago

Selected Answer: B

DynamoDB Accelerator, less over head.

upvoted 3 times

wmp7039 2 years, 4 months ago

Option B is incorrect as the constraint in the question is not to recode the application. DAX requires application to be reconfigured and point to DAX instead of DynamoDB

<https://docs.aws.amazon.com/amazondynamodb/latest/developerguide/DAX.client.modify-your-app.html>

Answer should be A

upvoted 3 times

LoXoL 1 year, 4 months ago

DAX does not require application logic modification (compatible with existing DynamoDB APIs). ElastiCache would work after changes on app's code.

upvoted 2 times

Buruguduystunstugudunstuy 2 years, 5 months ago

Selected Answer: B

To improve the performance efficiency of DynamoDB without reconfiguring the application, a solutions architect should recommend using Amazon DynamoDB Accelerator (DAX) which is Option B as the correct answer.

DAX is a fully managed, in-memory cache that can be used to improve the performance of read-intensive workloads on DynamoDB. DAX stores frequently accessed data in memory, allowing the application to retrieve data from the cache rather than making a request to DynamoDB. This can significantly reduce the number of read requests made to DynamoDB, improving the performance and reducing the latency of the application.

upvoted 4 times

Buruguduystunstugudunstuy 2 years, 5 months ago

Option A, using Amazon ElastiCache for Redis, would not be a good fit because it is not specifically designed for use with DynamoDB and would require reconfiguring the application to use it.

Option C, replicating data using DynamoDB global tables, would not directly improve the performance of reading requests and would require additional operational overhead to maintain the replication.

Option D, using Amazon ElastiCache for Memcached with Auto Discovery enabled, would also not be a good fit because it is not specifically designed for use with DynamoDB and would require reconfiguring the application to use it.

upvoted 2 times

career360guru 2 years, 5 months ago

Selected Answer: B

Option B

upvoted 3 times

k1kavi1 2 years, 6 months ago

Selected Answer: B

Agreed

upvoted 3 times

Shasha1 2 years, 6 months ago

B

DAX is a fully managed, highly available, in-memory cache for DynamoDB that delivers lightning-fast performance and consistent low-latency responses. It provides fast performance without requiring any application reconfiguration

upvoted 4 times

Wpcorgan 2 years, 6 months ago

B is correct

upvoted 2 times

goatbernard 2 years, 7 months ago

Selected Answer: B

DAX is the cache for this

upvoted 2 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

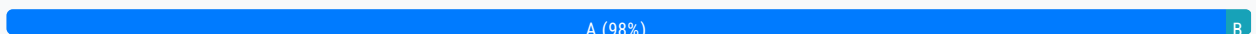
Question #178

Topic 1

A company's infrastructure consists of Amazon EC2 instances and an Amazon RDS DB instance in a single AWS Region. The company wants to back up its data in a separate Region.

Which solution will meet these requirements with the LEAST operational overhead?

- A. Use AWS Backup to copy EC2 backups and RDS backups to the separate Region. **Most Voted**
- B. Use Amazon Data Lifecycle Manager (Amazon DLM) to copy EC2 backups and RDS backups to the separate Region.
- C. Create Amazon Machine Images (AMIs) of the EC2 instances. Copy the AMIs to the separate Region. Create a read replica for the RDS DB instance in the separate Region.
- D. Create Amazon Elastic Block Store (Amazon EBS) snapshots. Copy the EBS snapshots to the separate Region. Create RDS snapshots. Export the RDS snapshots to Amazon S3. Configure S3 Cross-Region Replication (CRR) to the separate Region.

Correct Answer: A*Community vote distribution***Comments****cookieMr** **Highly Voted** 1 year, 11 months ago**Selected Answer: A**

Using AWS Backup to copy EC2 and RDS backups to the separate Region is the solution that meets the requirements with the least operational overhead. AWS Backup simplifies the backup process and automates the copying of backups to another Region, reducing the manual effort and operational complexity involved in managing separate backup processes for EC2 instances and RDS databases.

Option B is incorrect because Amazon Data Lifecycle Manager (Amazon DLM) is not designed for directly copying RDS backups to a separate region.

Option C is incorrect because creating Amazon Machine Images (AMIs) and read replicas adds complexity and operational overhead compared to a dedicated backup solution.

Option D is incorrect because using Amazon EBS snapshots, RDS snapshots, and S3 Cross-Region Replication (CRR) involves multiple manual steps and additional configuration, increasing complexity.

upvoted 12 times

vtbk Highly Voted 2 years, 5 months ago

Selected Answer: A

Cross-Region backup

Using AWS Backup, you can copy backups to multiple different AWS Regions on demand or automatically as part of a scheduled backup plan. Cross-Region backup is particularly valuable if you have business continuity or compliance requirements to store backups a minimum distance away from your production data.

<https://docs.aws.amazon.com/aws-backup/latest/devguide/whatisbackup.html>

upvoted 5 times

satyaamm Most Recent 4 months ago

Selected Answer: A

AWS Backups are the most suitable here.

upvoted 1 times

ChymKuBoy 11 months, 3 weeks ago

Selected Answer: A

A for sure

upvoted 2 times

thewalker 1 year, 4 months ago

Selected Answer: A

The easiest way to backup an EC2 instance and RDS Database would be to use AWS Backup. With AWS Backup you can:

Create a backup plan and select both the EC2 volume and RDS database for backup.

Choose a backup schedule that meets your requirements, such as daily or weekly backups.

AWS Backup will automatically take snapshots of the EC2 volume and backups of the RDS database as per the configured schedule.

The backups will be stored in S3 for long term retention based on your backup plan configuration.

upvoted 4 times

thewalker 1 year, 4 months ago

You can easily restore the EC2 volume or RDS database from these backups in case of data loss or corruption.

AWS manages the entire backup process so there is no operational overhead for you.

Some other options include using cron jobs to trigger snapshots and backups. But AWS Backup provides a fully managed service to centrally backup both EC2 and RDS with minimal effort.

The above is the output from Amazon Q.

upvoted 3 times

Guru4Cloud 1 year, 10 months ago

Selected Answer: A

AWS Backup provides a fully managed, centralized backup service across AWS services. It can be configured to automatically copy backups across Regions.

This requires minimal operational overhead compared to the other options:

upvoted 3 times

oguzbeliren 1 year, 10 months ago

D would have been a great option but the questions requires less manual effort. So, A is better.

upvoted 2 times

cheese929 2 years, 1 month ago

Selected Answer: A

A is correct

upvoted 3 times

kruasan 2 years, 1 month ago

Selected Answer: A

Option B, using Amazon Data Lifecycle Manager (Amazon DLM) to copy EC2 backups and RDS backups to the separate Region, would require more operational overhead because DLM is primarily designed for managing the lifecycle of Amazon EBS snapshots, and would require additional configuration to manage RDS backups.

Option C, creating AMLs of the EC2 instances and read replicas of the RDS DB instance in the separate Region, would require more manual effort to manage the backup and disaster recovery process, as it requires manual creation and management of AMLs and read replicas.

upvoted 4 times

kruasan 2 years, 1 month ago

Option D, creating EBS snapshots and RDS snapshots, exporting them to Amazon S3, and configuring S3 Cross-Region Replication (CRR) to the separate Region, would require more configuration and management effort. Additionally, S3 CRR can have additional charges for data transfer and storage in the destination region.

Therefore, option A is the best choice for meeting the company's requirements with the least operational overhead.

upvoted 4 times

gx2222 2 years, 2 months ago

Selected Answer: A

Option A, using AWS Backup to copy EC2 backups and RDS backups to the separate region, is the correct answer for the given scenario.

Using AWS Backup is a simple and efficient way to backup EC2 instances and RDS databases to a separate region. It requires minimal operational overhead and can be easily managed through the AWS Backup console or API. AWS Backup can also provide automated scheduling and retention management for backups, which can help ensure that backups are always available and up to date.

upvoted 4 times

dan80 2 years, 5 months ago

A is correct - you need to find a backup solution for EC2 and RDS. DLM doesn't work with RDS, only with snapshots.

upvoted 2 times

techhb 2 years, 5 months ago

Selected Answer: A

using Amazon DLM to copy EC2 backups and RDS backups to the separate region, is not a valid solution because Amazon DLM does not support backing up data across regions.

upvoted 2 times

Buruguduystunstugudunstuy 2 years, 5 months ago

Selected Answer: B

Option B. Use Amazon Data Lifecycle Manager (Amazon DLM) to copy EC2 backups and RDS backups to the separate Region.

Amazon DLM is a fully managed service that helps automate the creation and retention of Amazon EBS snapshots and RDS DB snapshots. It can be used to create and manage backup policies that specify when and how often snapshots should be created, as well as how long they should be retained. With Amazon DLM, you can easily and automatically create and manage backups of your EC2 instances and RDS DB instances in a separate Region, with minimal operational overhead.

upvoted 1 times

HayLLIHuK 2 years, 5 months ago

Buruguduystunstugudunstuy, sorry, but I haven't found any info about copying RDS backups by DLM. The DLM works only with EBS.

So the only answer is A - AWS Backup

upvoted 3 times

Buruguduystunstugudunstuy 2 years, 5 months ago

Option A, using AWS Backup to copy EC2 backups and RDS backups to the separate Region, would also work, but it may require more manual configuration and management.

Option C, creating AMLs of the EC2 instances and copying them to the separate Region, and creating a read replica for the

Option C, creating AMIs of the EC2 instances and copying them to the separate Region, and creating a read replica for the RDS DB instance in the separate Region, would work, but it may require more manual effort to set up and maintain.

Option D, creating EBS snapshots and copying them to the separate Region, creating RDS snapshots, and exporting them to Amazon S3, and configuring S3 CRR to the separate Region, would also work, but it would involve multiple steps and may require more manual effort to set up and maintain. Overall, using Amazon DLM is likely to be the easiest and most efficient option for meeting the requirements with the least operational overhead.

upvoted 1 times

PassNow1234 2 years, 5 months ago

Some of your answers are very detailed. Can you back them up with a reference?

upvoted 2 times

jwu413 2 years, 4 months ago

All of their answers are from ChatGPT

upvoted 5 times

techhb 2 years, 5 months ago

using Amazon DLM to copy EC2 backups and RDS backups to the separate region, is not a valid solution because Amazon DLM does not support backing up data across regions.

upvoted 5 times

egmiranda 2 years, 4 months ago

I choose A, but DLM support cross regions. DLM doesn't support RDS. Cross region copy rules it's a feature of DLM ("For each schedule, you can define the frequency, fast snapshot restore settings (snapshot lifecycle policies only), cross-Region copy rules, and tags")

<https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/snapshot-lifecycle.html>

upvoted 2 times

PassNow1234 2 years, 5 months ago

Thanks techhb

upvoted 2 times

Kruiz29 2 years, 5 months ago

This guy is giving wrong answers in detail...lol

upvoted 5 times

YogK 2 years ago

AWS DLM does not support RDS backups, only works with EBS storage.

<https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/snapshot-lifecycle.html>

upvoted 2 times

career360guru 2 years, 5 months ago

Selected Answer: A

Option A as it is fully managed service with least operational overhead

upvoted 2 times

Shasha1 2 years, 6 months ago

A

AWS Backup is a fully managed service that handles the process of copying backups to a separate Region automatically

upvoted 2 times

babaxoxo 2 years, 6 months ago

Selected Answer: A

Ans A with least operational overhead

upvoted 2 times

rjam 2 years, 6 months ago

AWS Backup supports cross-region backups

AWS Backup supports cross region backups
upvoted 4 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #179

Topic 1

A solutions architect needs to securely store a database user name and password that an application uses to access an Amazon RDS DB instance. The application that accesses the database runs on an Amazon EC2 instance. The solutions architect wants to create a secure parameter in AWS Systems Manager Parameter Store.

What should the solutions architect do to meet this requirement?

A. Create an IAM role that has read access to the Parameter Store parameter. Allow Decrypt access to an AWS Key Management Service (AWS KMS) key that is used to encrypt the parameter. Assign this IAM role to the EC2 instance.

Most Voted

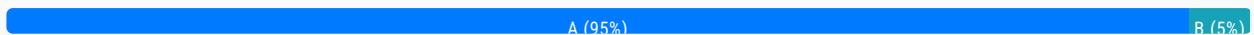
B. Create an IAM policy that allows read access to the Parameter Store parameter. Allow Decrypt access to an AWS Key Management Service (AWS KMS) key that is used to encrypt the parameter. Assign this IAM policy to the EC2 instance.

C. Create an IAM trust relationship between the Parameter Store parameter and the EC2 instance. Specify Amazon RDS as a principal in the trust policy.

D. Create an IAM trust relationship between the DB instance and the EC2 instance. Specify Systems Manager as a principal in the trust policy.

Correct Answer: A

Community vote distribution



Comments

Buruguduystunugudunstuy Highly Voted 1 year, 11 months ago

Selected Answer: A

CORRECT Option A

To securely store a database user name and password in AWS Systems Manager Parameter Store and allow an application running on an EC2 instance to access it, the solutions architect should create an IAM role that has read access to the Parameter Store parameter and allow Decrypt access to an AWS KMS key that is used to encrypt the parameter. The solutions architect should then assign this IAM role to the EC2 instance.

This approach allows the EC2 instance to access the parameter in the Parameter Store and decrypt it using the specified KMS key while enforcing the necessary security controls to ensure that the parameter is only accessible to authorized parties.

upvoted 25 times

Buruguduystunstugudunstuy 1 year, 11 months ago

Option B, would not be sufficient, as IAM policies cannot be directly attached to EC2 instances.

Option C, would not be a valid solution, as the Parameter Store parameter and the EC2 instance are not entities that can be related through an IAM trust relationship.

Option D, would not be a valid solution, as the trust policy would not allow the EC2 instance to access the parameter in the Parameter Store or decrypt it using the specified KMS key.

upvoted 9 times

sdasdawa **Highly Voted** 2 years ago

Selected Answer: A

Agree with A, IAM role is for services (EC2 for example)
IAM policy is more for users and groups

upvoted 10 times

lofzee **Most Recent** 6 months, 2 weeks ago

A all day. Don't even need to read the other answers.
You can't attach a policy to EC2. You have to attach a role.

upvoted 3 times

awsgeek75 11 months ago

Selected Answer: A

policy needs to be assigned to something so B is inaccurate
CD are just made up things

upvoted 3 times

TariqKipkemei 1 year, 3 months ago

Selected Answer: A

Create an IAM role that has read access to the Parameter Store parameter. Allow Decrypt access to an AWS Key Management Service (AWS KMS) key that is used to encrypt the parameter. Assign this IAM role to the EC2 instance

upvoted 2 times

Guru4Cloud 1 year, 3 months ago

Selected Answer: A

CORRECT Option A

upvoted 2 times

cookieMr 1 year, 5 months ago

Selected Answer: A

By creating an IAM role with read access to the Parameter Store parameter and Decrypt access to the associated AWS KMS key, the EC2 will have the necessary permissions to securely retrieve and decrypt the database user name and password from the Parameter Store. This approach ensures that the sensitive information is protected and can be accessed only by authorized entities.

Answers B, C, and D are not correct because they do not provide a secure way to store and retrieve the database user name and password from the Parameter Store. IAM policies, trust relationships, and associations with the DB instance are not the appropriate mechanisms for securely managing sensitive credentials in this scenario. Answer A is the correct choice as it involves creating an IAM role with the necessary permissions and assigning it to the EC2 instance to access the Parameter Store securely.

upvoted 3 times

cheese929 1 year, 7 months ago

Selected Answer: A

A is correct

upvoted 2 times

kruasan 1 year, 7 months ago

Selected Answer: A

By creating an IAM role and assigning it to the EC2 instance, the application running on the EC2 instance can access the Parameter Store parameter securely without the need for hard-coding the database user name and password in the application code.

The IAM role should have read access to the Parameter Store parameter and Decrypt access to an AWS KMS key that is used to encrypt the parameter to ensure that the parameter is protected at rest.

upvoted 2 times

HayLLIHuK 1 year, 11 months ago

There should be the Decrypt access to KMS.

"If you choose the SecureString parameter type when you create your parameter, Systems Manager uses AWS KMS to encrypt the parameter value."

<https://docs.aws.amazon.com/systems-manager/latest/userguide/systems-manager-parameter-store.html>

IAM role - for EC2

upvoted 2 times

BENICE 1 year, 11 months ago

A -- is correct option

upvoted 2 times

career360guru 1 year, 12 months ago

Option A.

upvoted 2 times

k1kavi1 1 year, 12 months ago

Selected Answer: A

A is correct

upvoted 2 times

Shasha1 2 years ago

Answer A

Create an IAM role that has read access to the Parameter Store parameter. Allow Decrypt access to an AWS Key Management Service (AWS KMS) key that is used to encrypt the parameter. Assign this IAM role to the EC2 instance. This solution will allow the application to securely access the database user name and password stored in the parameter store.

upvoted 2 times

[Removed] 2 years ago

Selected Answer: B

i think policy

upvoted 1 times

pentium75 11 months, 3 weeks ago

IAM Policies can be attached to IAM roles, and EC2 instances can be allowed to use IAM roles. You can't attach an IAM policy to an EC2 instance.

upvoted 2 times

[Removed] 2 years ago

Access to Parameter Store is enabled by IAM policies and supports resource level permissions for access. An IAM policy that grants permissions to specific parameters or a namespace can be used to limit access to these parameters. CloudTrail logs, if enabled for the service, record any attempt to access a parameter.

upvoted 1 times

[Removed] 2 years ago

<https://aws.amazon.com/blogs/compute/managing-secrets-for-amazon-ecs-applications-using-parameter-store-and-iam-roles-for-tasks/>

upvoted 1 times

JayBee65 1 year, 11 months ago

This link gives the example "Walkthrough: Securely access Parameter Store resources with IAM roles for tasks" - essentially A above. It does not show how this can be done using a policy (B) alone.

upvoted 2 times

turalmth 2 years ago

can you attach policy to ec2 directly ?

upvoted 2 times

EKA_CloudGod 2 years ago

Selected Answer: A

A. Attach IAM role to EC2 Instance

<https://aws.amazon.com/blogs/security/digital-signing-asymmetric-keys-aws-kms/>

upvoted 2 times

babaxoxo 2 years ago

Selected Answer: A

Attach IAM role to EC2 Instance profile

upvoted 4 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #180

Topic 1

A company is designing a cloud communications platform that is driven by APIs. The application is hosted on Amazon EC2 instances behind a Network Load Balancer (NLB). The company uses Amazon API Gateway to provide external users with access to the application through APIs. The company wants to protect the platform against web exploits like SQL injection and also wants to detect and mitigate large, sophisticated DDoS attacks.

Which combination of solutions provides the MOST protection? (Choose two.)

A. Use AWS WAF to protect the NLB.

B. Use AWS Shield Advanced with the NLB. **Most Voted**

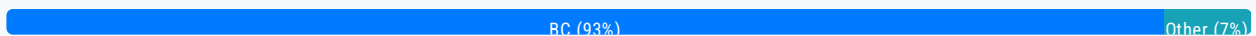
C. Use AWS WAF to protect Amazon API Gateway. **Most Voted**

D. Use Amazon GuardDuty with AWS Shield Standard

E. Use AWS Shield Standard with Amazon API Gateway.

Correct Answer: BC

Community vote distribution



Comments

babaxoxo **Highly Voted** 2 years ago

Selected Answer: BC

Shield - Load Balancer, CF, Route53
AWF - CF, ALB, API Gateway
upvoted 49 times

YogK 1 year, 6 months ago

Shield - Amazon Elastic Compute Cloud (EC2), Elastic Load Balancing (ELB), Amazon CloudFront, AWS Global Accelerator, and Route 53.

WAF - Amazon CloudFront, the Application Load Balancer (ALB), Amazon API Gateway, and AWS AppSync
upvoted 14 times

Ouk 1 year, 11 months ago

Thank u U meant WAF* - CloudFormation, right? haha
upvoted 5 times

rjam Highly Voted 2 years ago

Selected Answer: BC

AWS Shield Advanced - DDos attacks
AWS WAF to protect Amazon API Gateway, because WAF sits before the API Gateway and then comes NLB.
upvoted 8 times

studynoplay 1 year, 7 months ago

don't agree that NLB sits before API gateway. it should be other way around
upvoted 3 times

aadityaravi8 1 year, 5 months ago

yes.. coming from outside to inside... first of all DDos protection is required so the outer most NLB with Shield Advanced and then filter particular request doing SQL injection and all i.e API Gateway with WAF
upvoted 2 times

satyaamm Most Recent 4 months ago

Selected Answer: BC

AWS Shield protects against Layer 3 and 4 attacks through NLB and AWS WAF protects against Layer 7 attacks through API Gateway.
upvoted 1 times

Dharmarajan 4 months ago

Selected Answer: BC

I did not get this right initially, but when I looked it up, it became clear.
Basically any device that acts at network level - Shield (Layer 4 devices, maybe capable of layer 7 as well). For example, NLB, ELB, CF, Route 53.
Any device that works on HTTP/HTTPS/SFTP level(Layer 7 services) ==> WAF , which is ALB, API Gateway.

a Layer 7 device are not capable of layer 4 sevice, and they rely on underlying hardware/firmware(OR in this case of Software defined networking, Software) to do that.
upvoted 1 times

Dharmarajan 4 months ago

Definitely a very good question, to be noted for the exam.
upvoted 2 times

EMPERBACH 7 months, 3 weeks ago

Selected Answer: BD

B- (Shield Advance) PROTECT the platform against web exploits like SQL injection
D- (GuardDuty) also wants to DETECT mitigate large, sophisticated DDos attacks
WAF use for filter traffic, not make sense here.
upvoted 1 times

lofzee 6 months, 2 weeks ago

Shield advanced does not protect against SQL injection. That is what WAF is for.
GuardDuty is not the right tool here.
Answers are B and C bro.
upvoted 3 times

Guru4Cloud 1 year, 3 months ago

Selected Answer: BC

B) Use AWS Shield Advanced with the NLB

C) Use AWS WAF to protect Amazon API Gateway

The key reasons are:

AWS Shield Advanced provides expanded DDoS protection against larger and more sophisticated attacks
Using it with the NLB helps protect against network floods
WAF still provides critical protection against exploits at the API layer
upvoted 6 times

Sat897 1 year, 4 months ago

Selected Answer: BC

WAF - can't support NLB and it supports API Gateway
AWS Shield Advanced - NLB - DDOS
upvoted 2 times

cookieMr 1 year, 5 months ago

- B. AWS Shield Advanced provides advanced DDoS protection for the NLB, making it the appropriate choice for protecting against large and sophisticated DDoS attacks at the network layer.
- C. AWS WAF is designed to provide protection at the application layer, making it suitable for securing the API Gateway against web exploits like SQL injection.
- A. AWS WAF is not compatible with NLB as it operates at the application layer, whereas NLB operates at the transport layer.
- D. While GuardDuty helps detect threats, it does not directly protect against web exploits or DDoS attacks. Shield Standard focuses on edge resources, not specifically NLBs.
- E. Shield Standard provides basic DDoS protection for edge resources, but it does not directly protect the NLB or address web exploits at the application layer.
- upvoted 7 times

cheese929 1 year, 7 months ago

Selected Answer: BC

B and C is correct
upvoted 2 times

kruasan 1 year, 7 months ago

Selected Answer: BC

NLB is a Layer 3/4 component while WAF is a Layer 7 protection component.

That is why WAF is only available for Application Load Balancer in the ELB portfolio. NLB does not terminate the TLS session therefore WAF is not capable of acting on the content. I would consider using AWS Shield at Layer 3/4.
<https://repost.aws/questions/QU2fYXwSWUS0q9vZiWDoaEzA/nlb-need-to-attach-aws-waf>
upvoted 5 times

jdr75 1 year, 8 months ago

Selected Answer: C

- A. Use AWS WAF to protect the NLB.
INCORRECT, cos' WAF not integrate with network LB
- B. Use AWS Shield Advanced with the NLB.

YES. AWS Shield Advanced provides additional protections against more sophisticated and larger attacks for your applications running in AWS.
The doubt is : why apply the protection in the NLB when the facing of the app. is the API Gateway?, because Shield should be in front of the communications, not behind.
Nevertheless, this is the best option.

- C. Use AWS WAF to protect Amazon API Gateway.
YES, <https://aws.amazon.com/es/waf/faqs/>
 - D. Use Amazon GuardDuty with AWS Shield Standard
INCORRECT, GuardDuty not prevent attacks.
 - E. Use AWS Shield Standard with Amazon API Gateway.
INCORRECT. It could be, in principle, a good option, cos' it's in front of the gateway, but the questions said explicitly: "wants to detect and mitigate large, sophisticated DDoS attacks", and Standard not provide this feature.
- upvoted 2 times

kerl 1 year, 10 months ago

bullrem 1 year, 10 months ago

for those who select A, it is wrong, WAF is Layer 7, it only support ABL, APIGateway, CloudFront, COgnito User Pool and AppSync GraphQL API (<https://docs.aws.amazon.com/waf/latest/developerguide/waf-chapter.html>). NLB is NOT supported. Answer is BC
upvoted 4 times

bullrem 1 year, 10 months ago

Selected Answer: AB

A and B are the best options to provide the greatest protection for the platform against web vulnerabilities and large, sophisticated DDoS attacks.

Option A: Use AWS WAF to protect the NLB. This will provide protection against common web vulnerabilities such as SQL injection.

Option B: Use AWS Shield Advanced with the NLB. This will provide additional protection against large and sophisticated DDoS attacks.

upvoted 2 times

bullrem 1 year, 10 months ago

A and C are the best options for protecting the platform against web vulnerabilities and detecting and mitigating large and sophisticated DDoS attacks.

A: AWS WAF can be used to protect the NLB from web vulnerabilities such as SQL injection.

C: AWS WAF can be used to protect Amazon API Gateway and also provide protection against DDoS attacks.

B: AWS Shield Advanced is used to protect resources from DDoS attacks, but it is not specific to the NLB and may not provide the same level of protection as using WAF specifically on the NLB.

D and E: Amazon GuardDuty and AWS Shield Standard are primarily used for threat detection and may not provide the same level of protection as using WAF and Shield Advanced.

upvoted 1 times

Arifzefen 1 year, 5 months ago

A is not correct as WAF doesn't support Network Load Balancer

upvoted 2 times

omoakin 1 year, 6 months ago

correct

upvoted 1 times

bullrem 1 year, 10 months ago

The best protection for the platform would be to use A and C together because it will protect both the NLB and the API Gateway from web vulnerabilities and DDoS attacks.

upvoted 1 times

drabi 1 year, 11 months ago

Selected Answer: BC

AWS Shield Advanced can help protect your Amazon EC2 instances and Network Load Balancers against infrastructure-layer Distributed Denial of Service (DDoS) attacks. Enable AWS Shield Advanced on an AWS Elastic IP address and attach the address to an internet-facing EC2 instance or Network Load Balancer. <https://aws.amazon.com/blogs/security/tag/network-load-balancers/>

upvoted 2 times

duriselvan 1 year, 11 months ago

Regional resources

You can protect regional resources in all Regions where AWS WAF is available. You can see the list at AWS WAF endpoints and quotas in the Amazon Web Services General Reference.

You can use AWS WAF to protect the following regional resource types:

Amazon API Gateway REST API

Application Load Balancer

AWS AppSync GraphQL API

Amazon Cognito user pool

You can only associate a web ACL to an Application Load Balancer that's within AWS Regions. For example, you cannot associate a web ACL to an Application Load Balancer that's on AWS Outposts.

upvoted 1 times

duriselvan 1 year, 11 months ago

Ans:-a and C

upvoted 1 times

Buruguduystunstugudunstuy 1 year, 11 months ago

Selected Answer: AC

CORRECT

A. Use AWS WAF to protect the NLB.

C. Use AWS WAF to protect Amazon API Gateway.

AWS WAF is a web application firewall that helps protect web applications from common web exploits such as SQL injection and cross-site scripting attacks. By using AWS WAF to protect the NLB and Amazon API Gateway, the company can provide an additional layer of protection for its cloud communications platform against these types of web exploits.

upvoted 1 times

Buruguduystunstugudunstuy 1 year, 11 months ago

About AWS Shield Advanced and Amazon GuardDuty

AWS Shield Advanced is a managed DDoS protection service that provides additional protection for Amazon EC2 instances, Amazon RDS DB instances, Amazon Elastic Load Balancers, and Amazon CloudFront distributions. It can help detect and mitigate large, sophisticated DDoS attacks, "but it does not provide protection against web exploits like SQL injection."

Amazon GuardDuty is a threat detection service that uses machine learning and other techniques to identify potentially malicious activity in your AWS accounts. It can be used in conjunction with AWS Shield Standard, which provides basic DDoS protection for Amazon EC2 instances, Amazon RDS DB instances, and Amazon Elastic Load Balancers. However, neither Amazon GuardDuty nor AWS Shield Standard provides protection against web exploits like SQL injection.

Overall, the combination of using AWS WAF to protect the NLB and Amazon API Gateway provides the most protection against web exploits and large, sophisticated DDoS attacks.

upvoted 1 times

PassNow1234 1 year, 11 months ago

Your answer is wrong.

Sophisticated DDOS = Shield Advanced (DDOS attacks the front!) What happens if your load balances goes down?

Your API gateway is on the BACK further behind the NLB. SQL Protect that with the WAF

B and C are right.

upvoted 5 times

jwu413 1 year, 10 months ago

This guy just copies and pastes from ChatGPT.

upvoted 5 times

BENICE 1 year, 11 months ago

Option B and C

upvoted 1 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #181

Topic 1

A company has a legacy data processing application that runs on Amazon EC2 instances. Data is processed sequentially, but the order of results does not matter. The application uses a monolithic architecture. The only way that the company can scale the application to meet increased demand is to increase the size of the instances.

The company's developers have decided to rewrite the application to use a microservices architecture on Amazon Elastic Container Service (Amazon ECS).

What should a solutions architect recommend for communication between the microservices?

- A. Create an Amazon Simple Queue Service (Amazon SQS) queue. Add code to the data producers, and send data to the queue. Add code to the data consumers to process data from the queue. **Most Voted**
- B. Create an Amazon Simple Notification Service (Amazon SNS) topic. Add code to the data producers, and publish notifications to the topic. Add code to the data consumers to subscribe to the topic.
- C. Create an AWS Lambda function to pass messages. Add code to the data producers to call the Lambda function with a data object. Add code to the data consumers to receive a data object that is passed from the Lambda function.
- D. Create an Amazon DynamoDB table. Enable DynamoDB Streams. Add code to the data producers to insert data into the table. Add code to the data consumers to use the DynamoDB Streams API to detect new table entries and retrieve the data.

Correct Answer: A*Community vote distribution*

A (95%)

B (5%)

Comments**Buruguduystunstugudunstuy** **Highly Voted** 1 year, 11 months ago**Selected Answer: A**

Option B, using Amazon Simple Notification Service (SNS), would not be suitable for this use case, as SNS is a pub/sub messaging service that is designed for one-to-many communication, rather than point-to-point communication between specific microservices.

Option C, using an AWS Lambda function to pass messages, would not be suitable for this use case, as it would require the data producers and data consumers to have a direct connection and invoke the Lambda function, rather than being decoupled through a message queue.

Option D, using an Amazon DynamoDB table with DynamoDB Streams, would not be suitable for this use case, as it would require the data consumers to continuously poll the DynamoDB Streams API to detect new table entries, rather than being notified of new data through a message queue.

upvoted 21 times

jerryI 2 months, 3 weeks ago

I think A is obvious the most suitable, but i dont understand how is this related to ECS mentioned in the question

upvoted 1 times

Buruguduystunstugudunstuy 1 year, 11 months ago

Hence, Option A is the correct answer.

Create an Amazon Simple Queue Service (Amazon SQS) queue. Add code to the data producers, and send data to the queue. Add code to the data consumers to process data from the queue.

upvoted 10 times

cookieMr Highly Voted 1 year, 5 months ago

Selected Answer: A

A. Creating an Amazon SQS queue allows for asynchronous communication between microservices, decoupling the data producers and consumers. It provides scalability, flexibility, and ensures that data processing can happen independently and at a desired pace.

B. Amazon SNS is more suitable for pub/sub messaging, where multiple subscribers receive the same message. It may not be the best fit for sequential data processing.

C. Using AWS Lambda functions for communication introduces unnecessary complexity and may not be the optimal solution for sequential data processing.

D. Amazon DynamoDB with DynamoDB Streams is primarily designed for real-time data streaming and change capture scenarios. It may not be the most efficient choice for sequential data processing in a microservices architecture.

upvoted 7 times

satyaamm Most Recent 4 months ago

Selected Answer: A

SQS Queues are most suitable as they provide decoupling between microservices.

upvoted 1 times

scar0909 9 months, 1 week ago

Selected Answer: A

A for sure

upvoted 2 times

reviewmine 9 months, 4 weeks ago

Selected Answer: A

To Decouple a monolithic application - SQS

- SQS standard - not in order

- SQS FIFO - in order

upvoted 5 times

upliftinghut 10 months, 3 weeks ago

Selected Answer: A

Data is processed sequentially, but the order of results does not matter => SQS; if order matters => SQL FIFO

upvoted 3 times

Cloud_A 11 months, 1 week ago

Selected Answer: A

A is the answer.

upvoted 2 times

TariqKipkemei 1 year, 3 months ago

Selected Answer: A

Data is processed sequentially, but the order of results does not matter = Amazon Simple Queue Service
upvoted 2 times

Guru4Cloud 1 year, 3 months ago

Selected Answer: A

A) Create an Amazon Simple Queue Service (Amazon SQS) queue. Add code to the data producers, and send data to the queue. Add code to the data consumers to process data from the queue.

For asynchronous communication between decoupled microservices, an SQS queue is the most appropriate service to use.

SQS provides a scalable, highly available queue to buffer messages between producers and consumers.
The order of processing does not matter, so a queue model fits well.
The consumers can scale independently to process messages from the queue.

upvoted 4 times

omoakin 1 year, 6 months ago

BBBBBBBBBB

upvoted 1 times

Bmarodi 1 year, 6 months ago

Selected Answer: A

SQS for decoupling a monolithic architecture, hence option A is the right answer.

upvoted 2 times

Madhuaws 1 year, 8 months ago

it also says 'the order of results does not matter'. Option B is correct.

upvoted 1 times

asoli 1 year, 8 months ago

Selected Answer: A

The answer is A.
B is wrong because SNS cannot send events "directly" to ECS.
<https://docs.aws.amazon.com/sns/latest/dg/sns-event-destinations.html>

upvoted 2 times

user_deleted 1 year, 9 months ago

Selected Answer: B

it doesn't say it is one-one relationships, SNS is better

upvoted 3 times

markw92 1 year, 5 months ago

watch out for this sentence in the question..."Data needs to process sequentially...."

upvoted 3 times

career360guru 1 year, 12 months ago

Selected Answer: A

Best answer is A.
Though C or D is possible it requires additional components and integration and so they are not efficient. Assuming that rate of incoming requests is within limits that SQS can handle A is best option.

upvoted 2 times

k1kavi1 1 year, 12 months ago

Selected Answer: A

A is correct

upvoted 2 times

Shasha1 2 years ago

answer is B.

An Amazon Simple Notification Service (Amazon SNS) topic can be used for communication between the microservices in this scenario. The data producers can be configured to publish notifications to the topic, and the data consumers can be configured to subscribe to the topic and receive notifications as they are published. This allows for asynchronous communication between the microservices, Question here focus on communication between microservices

upvoted 2 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #182

Topic 1

A company wants to migrate its MySQL database from on premises to AWS. The company recently experienced a database outage that significantly impacted the business. To ensure this does not happen again, the company wants a reliable database solution on AWS that minimizes data loss and stores every transaction on at least two nodes.

Which solution meets these requirements?

- A. Create an Amazon RDS DB instance with synchronous replication to three nodes in three Availability Zones.
- B. Create an Amazon RDS MySQL DB instance with Multi-AZ functionality enabled to synchronously replicate the data.
Most Voted
- C. Create an Amazon RDS MySQL DB instance and then create a read replica in a separate AWS Region that synchronously replicates the data.
- D. Create an Amazon EC2 instance with a MySQL engine installed that triggers an AWS Lambda function to synchronously replicate the data to an Amazon RDS MySQL DB instance.

Correct Answer: B*Community vote distribution*

B (98%)

C

Comments**rjam** **Highly Voted** 2 years, 6 months ago**Selected Answer: B**

Amazon RDS MySQL DB instance with Multi-AZ functionality enabled to synchronously replicate the data
Standby DB in Multi-AZ- synchronous replication

Read Replica always asynchronous. so option C is ignored.

upvoted 20 times

studynoplay **Highly Voted** 2 years, 1 month ago**Selected Answer: B**

RDS Multi-AZ = Synchronous = Disaster Recovery (DR)

Read Replica = Asynchronous = High Availability

upvoted 15 times

pentium75 1 year, 5 months ago

B is correct but the explanation is flawed ;)

RDS Multi-AZ = Synchronous = High Availability
Read Replica = Asynchronous = Disaster Recovery (DR)
upvoted 7 times

satyaamm Most Recent 4 months ago

Selected Answer: B

RDS Multi AZ is the most suitable here.
upvoted 1 times

PoolDead 10 months, 1 week ago

"Minimizes Data Loss" Therefore answer is B.

Amazon RDS read replicas use asynchronous replication, not synchronous. Therefore, this option does not meet the requirement for minimizing data loss as asynchronous replication can result in data lag.
upvoted 2 times

Nawaff 11 months, 1 week ago

Selected Answer: B

Answer is B
Find the below URL for the perfect explanation for the differences between:
- Multi-AZ DB
- Multi-Region DB
- Read replicas DB

<https://aws.amazon.com/rds/features/read-replicas/>
upvoted 2 times

scar0909 1 year, 3 months ago

Selected Answer: B

Multi AZ for availability
upvoted 2 times

riyasara 1 year, 7 months ago

Option A is incorrect because Amazon RDS does not support synchronous replication to three nodes in three Availability Zones. Option C is incorrect because while you can create a read replica in a separate AWS Region, the replication from the primary DB instance to the read replica is asynchronous, not synchronous.
upvoted 4 times

cookieMr 1 year, 11 months ago

Selected Answer: B

B. Create an Amazon RDS MySQL DB instance with Multi-AZ functionality enabled to synchronously replicate the data.

Enabling Multi-AZ functionality in Amazon RDS ensures synchronous replication of data to a standby replica in a different Availability Zone. This provides high availability and minimizes data loss in the event of a database outage.

A. Creating an Amazon RDS DB instance with synchronous replication to three nodes in three Availability Zones would provide even higher availability but is not necessary for the stated requirements.

C. Creating a read replica in a separate AWS Region would provide disaster recovery capabilities but does not ensure synchronous replication or meet the requirement of storing every transaction on at least two nodes.

D. Using an EC2 instance with a MySQL engine and triggering an AWS Lambda function for replication introduces unnecessary complexity and is not the most suitable solution for ensuring reliable and synchronous replication.
upvoted 3 times

channn 2 years, 2 months ago

Selected Answer: B

B

—
since all other answers r wrong
upvoted 3 times

jayce5 2 years, 2 months ago

Selected Answer: B

B
Since read replica is async.
upvoted 2 times

LuckyAro 2 years, 4 months ago

Selected Answer: C

Multi AZ is not as protected as Multi-Region Read Replica.
upvoted 1 times

pentium75 1 year, 5 months ago

But is IS protected. Read replica is asynchronous, fails to meet the "store EVERY transaction on at least two nodes" requirement.
upvoted 1 times

JayBee65 2 years, 5 months ago

I curios to know why A isn't right. Is it just that it would take more effort?
upvoted 3 times

pentium75 1 year, 5 months ago

How would you implement A?
upvoted 1 times

techhb 2 years, 5 months ago

B is correct C requires more wokr.
upvoted 2 times

BENICE 2 years, 5 months ago

Option B
upvoted 2 times

bammy 2 years, 5 months ago

Multi-AZ will give at least two nodes as required by the question. The answer is B.

Amazon RDS provides high availability and failover support for DB instances using Multi-AZ deployments with a single standby DB instance.
<https://docs.aws.amazon.com/AmazonRDS/latest/UserGuide/Concepts.MultiAZSingleStandby.html>
upvoted 5 times

career360guru 2 years, 5 months ago

Selected Answer: B

Option B
upvoted 2 times

Shasha1 2 years, 6 months ago

Option A is the correct answer in this scenario because it meets the requirements specified in the question. It creates an Amazon RDS DB instance with synchronous replication to three nodes in three Availability Zones, which will provide high availability and durability for the database, ensuring that the data is stored on multiple nodes and automatically replicated across Availability Zones.

Option B is not a correct answer because it creates an Amazon RDS MySQL DB instance with Multi-AZ functionality enabled, which only provides failover capabilities. It does not enable synchronous replication to multiple nodes, which is required in this scenario.
upvoted 2 times